INCH-POUND

MIL-M-38510/103E
17 December 1993
SUPERSEDING
MIL-M-38510/103D
24 June 1988

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, VOLTAGE COMPARATORS,

MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, voltage comparators. Two product assurance classes and a choice of case outline and lead finish are provided for each type and are reflected in the complete Part or Identifying Number (PIN). The term Part or Identifying Number (PIN) is equivalent to the term part number which was previously used in this specification.

1.2 PIN. The complete PIN shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as shown in the following:

Device type	Circuit
01	Single differential voltage comparator
02	Dual channel differential voltage comparator
03	Single voltage comparator/buffer
04	Precision voltage comparator/buffer
05	Dual precision voltage comparator/buffer 1/
06	Dual precision high speed voltage comparator
07	Dual high precision, high speed voltage comparator

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated in MIL-STD-1835 and as follows:

Case outline	Descriptive designator	Terminals	Package style
A	GDFP5-F14 or CDFP6-F14	14	Flat pack
C	GDIP1-T14 or CDIP2-T14	14	Dual-in-line
E	GDIP1-T16 or CDIP2-T16	16	Dual-in-line
F	GDFP2-F16 or CDFP3-F16	16	Flat pack
G	MACY1-X8	8	Can
H	GDFP1-F10 or CDFP2-F10	10	Flat pack
I	MACY1-X10	10	Can
P	GDIP1-T8 or CDIP2-T8	8	Dual-in-line
2	CQCC1-N20	20	Square leadless chip carrier

1/ Device type 05 may be monolithic, or it may consist of two separate, independent dice.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RADC/RBE-2), Griffiss AFB, NY 13441-5700, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5962

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1.3 Absolute maximum ratings.

	Device type 01	Device type 02	Device type 03	Device types 04 and 05	Device types 06 and 07
Positive supply voltage	+14.0 V	+14.0 V	+15.0 V	+30.0 V	+18 V
Negative supply voltage	-7.0 V	-7.0 V	-15.0 V	-30.0 V	-25 V
Total supply voltage	---	---	---	+36.0 V	+36 V
Output voltage	---	---	+24.0 V	---	---
Output to negative supply voltage	---	---	+30.0 V	+50.0 V	+36.0 V
Input voltage range	±7.0 V	±7.0 V	±7.0 V	2/	±15.0 V 3/
Differential input voltage	±5.0 V	±5.0 V	±5.0 V	±30.0 V	±5.0 V
Peak output current	10 mA	50 mA	---	---	25 mA
Sink current	---	---	100 mA	50 mA	---
Output short-circuit duration	10 s	10 s	10 s	10 s	10 s
Strobe voltage	---	6.0 V	6.0 V	---	---
Maximum strobe current	---	---	---	10 mA	---
Storage temperature range	-65°C to +150°C for all device types				
Junction temperature (T_J) 4/	+175°C	+175°C	+175°C	+175°C	+175°C
Lead temperature (soldering, 60 seconds)	+300°C	+300°C	+300°C	+300°C	+300°C

1.4 Recommended operating conditions.

Supply voltage range 3/:
- Device types 01 and 02 - $+V_{CC}$ = +12.0 V dc, $-V_{CC}$ = -6.0 V dc
- Device type 03 - $+V_{CC}$ = +12.0 V dc, $-V_{CC}$ = -3.0 to -12.0 V dc
- Device types 04, 05, 06, and 07 - $\pm V_{CC}$ = ±15.0 V dc

Ambient operating temperature range (T_A) - -55°C to +125°C

1.5 Power and thermal characteristics.

Case outline	Package	Maximum allowable power dissipation	Maximum θ_{JC}	Maximum θ_{JA}
A	14-lead flat package	350 mW at T_A = +125°C	60°C/W	140°C/W
C, E, and P	dual-in-line packages	400 mW at T_A = +125°C	35°C/W	120°C/W
G	8-lead can	330 mW at T_A = +125°C	40°C/W	150°C/W
I	10-lead can	350 mW at T_A = +125°C	40°C/W	140°C/W
H	10-lead flat package	330 mW at T_A = +125°C	60°C/W	150°C/W
F	16-lead flat package	350 mW at T_A = +125°C	60°C/W	140°C/W
2	20-terminal square chip carrier package	199 mW at T_A = +125°C	55°C/W	121°C/W

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

2/ The positive input voltage limit is 30 V above the negative supply. The negative input voltage limit is equal to the negative supply voltage or 30 V below the positive supply, whichever is less negative.
3/ For supply voltages less than ±15.0 V, the input voltage rating is equal to the supply voltage.
4/ For short term test (in the specific burn-in and life test configuration when required and up to 168 hours maximum) T_J = +275°C.

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SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.
MIL-STD-1835 - Microcircuit Case Outlines.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 40, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document shall takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Item requirements. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and as specified herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturer's schematics shall be maintained by the agent activity and will be available upon request.

3.2.3 Case outlines. The case outlines shall be in accordance with 1.2.3 herein.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510.

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I and apply over the full operating ambient temperature range of -55°C to +125°C.

3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

3.6 Electrical test requirements. Electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-M-38510.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 50 (see MIL-M-38510, appendix E).

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TABLE I. Electrical performance characteristics for device types 01, 02, and 03.

Characteristic	Symbol	Conditions 1/		Device type	Limits		Unit
					Min	Max	
Input offset voltage	V_{IO}	R_S = 200Ω and 50Ω	V_{OUT} = 1.4 V, T_A = +25°C	01	-2	+2	mV
				02	-3.5	+3.5	mV
			V_{OUT} = 1.8 V, T_A = -55°C	01	-3	+3	mV
				02	-4.5	+4.5	mV
			V_{OUT} = 1.0 V, T_A = +125°C	01	-3	+3	mV
				02	-4.5	+4.5	mV
			V_{OUT} = 1.5 V, T_A = +25°C	03	-2	+2	mV
			V_{OUT} = 1.5 V, -55°C ≤ T_A ≤ +125°C	03	-3	+3	mV
Input offset voltage temperature coefficient	$\Delta V_{IO}/\Delta T$	R_S = 50Ω	-55°C ≤ T_A ≤ +125°C	01,02,03	-10	+10	µV/°C
Input offset current	I_{IO}		V_{OUT} = 1.4 V, T_A = +25°C	01	-3	+3	µA
				02	-10	+10	µA
			V_{OUT} = 1.8 V, T_A = -55°C	01	-7	+7	µA
				02	-20	+20	µA
			V_{OUT} = 1.0 V, T_A = +125°C	01	-3	+3	µA
				02	-10	+10	µA
		V_{OUT} = 1.5 V	T_A = +25°C, +125°C	03	-3	+3	µA
			T_A = -55°C	03	-7	+7	µA
Input offset current temperature coefficient	$\Delta I_{IO}/\Delta T$		T_A = +125°C	01,02,03	-25	+25	nA/°C
			T_A = -55°C	01,02,03	-75	+75	nA/°C
Input bias current	$+I_{IB}$		T_A = +25°C, +125°C	01, 03	-.1	+20	µA
				02	-.1	+75	µA
			T_A = -55°C	01, 03	-.1	+45	µA
				02	-.1	+150	µA

See footnotes at end of device types 01, 02, and 03.

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TABLE I. Electrical performance characteristics for device types 01, 02, and 03 - Continued.

Characteristic	Symbol	Conditions 1/		Device type	Limits		Unit
					Min	Max	
Strobe current	I_{STROBE}	V_{STROBE} = 100 mV, V_{ID} = +10 mV		02	-2.5	-0.1	mA
		V_{STROBE} = 400 mV, V_{ID} = -5 mV		03	-3.3	-0.1	mA
Input voltage common mode rejection	CMR	R_S = 200Ω, -5 V ≤ V_{IN} ≤ +5 V, $-V_{CC}$ = -7.0 V		01,02,03	+80		dB
High level output voltage	V_{OH}	I_{OH} = 0 mA	V_{ID} = +5 mV	01	+2.5	+5.0	V
			V_{ID} = +10 mV	02			
		I_{OH} = -5 mA	V_{ID} = +5 mV	01	+2.5	+5.0	V
			V_{ID} = +10 mV	02			
		I_{OH} = -400 µA	V_{ID} = +5 mV	03	+2.5	+5.5	V
Low level output voltage	V_{OL}	I_{OL} = 0 mA	V_{ID} = -5 mV	01	-1.0	0	V
			V_{ID} = -10 mV	02			
		V_{ID} = -5 mV	T_A = +25°C, I_{OL} = 100 mA	03		+1.5	V
			T_A = -55°C, I_{OL} = 50 mA	03		+1.0	V
			T_A = +125°C, I_{OL} = 16 mA	03		+0.4	V
Strobed output level	V_O (strobed)	V_{STROBE} = 0.3 V, V_{ID} = +10 mV 2/		02	-1.0	0	V
		V_{ID} = -5 mV	V_{STROBE} = 0.9 V	03	+2.5	+5.5	V
			V_{STROBE} = 2.5 V, I_{OL} = 16 mA	03		+0.4	V

See footnotes at end of device types 01, 02, and 03.

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TABLE I. Electrical performance characteristics for device types 01, 02, and 03 - Continued.

Characteristic	Symbol	Conditions 1/			Device type	Limits		Unit
						Min	Max	
Output sink current	I_{OL}	V_{OL} = 0 V	T_A = +25°C	V_{ID} = -5 mV	01	+2.0		mA
				V_{ID} = -10 mV	02	+0.5		mA
			T_A = -55°C	V_{ID} = -5 mV	01	+1.0		mA
				V_{ID} = -10 mV	02	+0.25		mA
			T_A = +125°C	V_{ID} = -5 mV	01	+0.5		mA
				V_{ID} = -10 mV	02	+0.12		mA
Output leakage current	I_{CEX}	V_{OUT} = 24 V V_{ID} = +5 mV		T_A = +25°C	03	-.01	+1.0	µA
				T_A = -55°C	03	-0.5	+100	µA
				T_A = +125°C	03	-0.5	+100	µA
Response time, output saturated high level to threshold level	t_{HTHR}	See figure 6, T_A = +25°C, C_L = 5 pF, 100 mV step, 5 mV overdrive			01, 03		60	ns
					02		90	ns
Response time, output saturated low level to threshold level	t_{LTHR}	See figure 6, T_A = +25°C, C_L = 5 pF, 100 mV step, 5 mV overdrive			01,02,03		60	ns
Strobe release time	t_{STRL}	See figure 7, T_A = +25°C, C_L = 5 pF, V_{ID} = +10 mV			02, 03		15	ns
Positive supply current	$+I_{CC}$	V_{OUT} = V_{OL}	V_{ID} = -5 mV		01	+0.5	+9.0	mA
			V_{ID} = -10 mV		02	+0.5	+13.5	mA
		V_{ID} = -5 mV			03	+0.5	+10	mA

See footnotes at end of device types 01, 02, and 03.

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TABLE I. Electrical performance characteristics for device types 01, 02, and 03 - Continued.

Characteristic	Symbol	Conditions 1/		Device type	Limits		Unit
					Min	Max	
Negative supply current	$-I_{CC}$	$V_{OUT} = V_{OL}$ (no load)	V_{ID} = -5 mV	01	-7.0	-0.5	mA
			V_{ID} = -10 mV	02	-6.2	-0.5	mA
		V_{ID} = -5 mV		03	-3.6	-0.5	mA
Voltage gain	$A_V(\pm)$	ΔV_{OUT} = ±0.5 V	V_{OUT} = 1.4 V, T_A = +25°C	01	+1,250		V/V
				02	+750		V/V
			V_{OUT} = 1.8 V, T_A = -55°C	01	+1,000		V/V
				02	+500		V/V
			V_{OUT} = 1.0 V, T_A = +125°C	01	+1,000		V/V
				02	+500		V/V
Voltage gain	A_V+	ΔV_{OUT} = +3.0 V, V_{OUT} = 1.5 V	T_A = +25°C	03	+30		V/mV
			T_A = -55°C	03	+50		V/mV
			T_A = +125°C	03	+10		V/mV
Voltage gain	A_V-	ΔV_{OUT} = -1.0 V, V_{OUT} = 1.5 V	T_A = +25°C	03	+30		V/mV
			T_A = -55°C	03	+50		V/mV
			T_A = +125°C	03	+10		V/mV

1/ Unless otherwise specified, $+V_{CC}$ = +12 V dc and $-V_{CC}$ = -6 V dc. Unless otherwise specified, for device type 02, strobe on device not being tested is connected to ground; strobe on device being tested is left open. Unless otherwise specified, for device type 03, strobes are at a high level for all tests.

2/ The output voltage follows the strobe voltage, staying one diode drop (0.7 V) below.

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TABLE I. Electrical performance characteristics for device types 04 and 05.

Characteristic	Symbol	Conditions 1/		Limits		Unit
				Min	Max	
Input offset voltage	V_{IO}	R_S = 50Ω, V_{IC} = 0 V, 13 V and -14.5 V 2/	T_A = +25°C	-3.0	+3.0	mV
			-55°C ≤ T_A ≤ +125°C	-4.0	+4.0	mV
		R_S = 50Ω, V_{IC} = 0 V, $\pm V_{CC}$ = ±2.5 V	T_A = +25°C	-3.0	+3.0	mV
			-55°C ≤ T_A ≤ +125°C	-4.0	+4.0	mV
Raised input offset voltage 3/	$V_{IO(R)}$	R_S = 50Ω, V_{IC} = 0 V, 13 V, and -14.5 V 2/ V_{BAL} = $V_{BAL/STB}$ = $+V_{CC}$	T_A = +25°C	-3.0	+3.0	mV
			-55°C ≤ T_A ≤ +125°C	-4.5	+4.5	mV
Input offset voltage temperature coefficient	$\Delta V_{IO}/\Delta T$	R_S = 50Ω		-25	+25	µV/°C
Input offset current	I_{IO}	R_S = 50 kΩ, V_{IC} = 0 V, 13 V, and -14.5 V 2/	+25°C ≤ T_A ≤ +125°C	-10	+10	nA
			T_A = -55°C	-20	+20	nA
Raised input offset current 3/	$I_{IO(R)}$	R_S = 50 kΩ, V_{IC} = 0 V, V_{BAL} = $V_{BAL/STB}$ = $+V_{CC}$	+25°C ≤ T_A ≤ +125°C	-25	+25	nA
			T_A = -55°C	-50	+50	nA
Input offset current temperature coefficient	$\Delta I_{IO}/\Delta T$	R_S = 50 kΩ	T_A = +125°C	-100	+100	pA/°C
			T_A = -55°C	-200	+200	pA/°C
Input bias current	$+I_{IB}$	R_S = 50 kΩ, V_{IC} = 0 V	+25°C ≤ T_A ≤ +125°C	-100	+0.1	nA
			T_A = -55°C	-150	+0.1	nA
		R_S = 50 kΩ, V_{IC} = 13 V and -14.5 V	+25°C ≤ T_A ≤ +125°C	-150	+0.1	nA
			T_A = -55°C	-200	+0.1	nA

See footnotes at the end of device types 04 and 05.

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TABLE I. Electrical performance characteristics for device types 04 and 05 - Continued.

Characteristic	Symbol	Conditions 1/		Limits		Unit
				Min	Max	
Input bias current	$-I_{IB}$	R_S = 50 kΩ, V_{IC} = 0 V	+25°C ≤ T_A ≤ +125°C	-100	+0.1	nA
			T_A = -55°C	-150	+0.1	nA
		R_S = 50 kΩ, V_{IC} = 13 V and -14.5 V	+25°C ≤ T_A ≤ +125°C	-150	+0.1	nA
			T_A = -55°C	-200	+0.1	nA
Collector output voltage (strobed)	$V_{O(STB)}$	R_S = 50Ω, I_{STB} = -3.0 mA		+14		V
Common mode rejection ratio	CMRR	R_S = 50Ω, V_{IC} = 13 V and -14.5 V		+80		dB
Output leakage current	I_{CEX}	$\pm V_{CC}$ = ±18 V, V_{OUT} = 32 V	T_A = +25°C	-1	+10	nA
			T_A = +125°C	-1	+500	nA
Input leakage current	I_{I1}	$\pm V_{CC}$ = ±18 V, V_{ID} = +29 V		-5	+500	nA
	I_{I2}	$\pm V_{CC}$ = ±18 V, V_{ID} = -29 V		-5	+500	nA
Positive supply current (limit is for one comparator of device type 05)	$+I_{CC}$		T_A = -55°C		+7.0	mA
			T_A = +25°C		+6.0	mA
			T_A = +125°C		+6.0	mA
Negative supply current (limit is for one comparator of device type 05)	$-I_{CC}$		T_A = -55°C	-6.0		mA
			T_A = +25°C	-5.0		mA
			T_A = +125°C	-5.0		mA

See footnotes at end of device types 04 and 05.

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partial penetration velocity being lower than the complete penetration

TABLE I. Electrical performance characteristics for device types 04 and 05 - Continued.

Characteristic	Symbol	Conditions 1/		Limits		Unit
				Min	Max	
Output short-circuit current	I_{OS}	10 ms maximum test duration	T_A = -55°C		+250	mA
			T_A = +25°C		+200	mA
			T_A = +125°C		+150	mA
Adjustment for input offset voltage	$V_{IO(ADJ)+}$	R_S = 50Ω	T_A = +25°C	+5.0		mV
	$V_{IO(ADJ)-}$	R_S = 50Ω	T_A = +25°C		-5.0	mV
Low level output voltage	V_{OL1} V_{OL2}	$+V_{CC}$ = +4.5 V, $-V_{CC}$ = 0 V, V_{IC} = -1.75 V, +0.75 V, V_{ID} = -6.0 mV, I_O = 8 mA			+0.4	V
	V_{OL3} V_{OL4}	$\pm V_{CC}$ = ±15 V, V_{ID} = -5.0 mV, V_{IC} = 13 V, -14 V, I_O = 50 mA			+1.5	V
Voltage gain (emitter follower output)	$\pm A_{VE}$	R_L = 600Ω	T_A = +25°C	+10		V/mV
			-55°C ≤ T_A ≤ +125°C	+8		V/mV
Response time, low-to-high level collector output	t_{RLHC}	V_{OD} (overdrive) = -5 mV, C_L = 50 pF, V_{IN} = 100 mV	-55°C ≤ T_A ≤ +25°C		300	ns
			T_A = +125°C		640	ns

See footnotes at end of device types 04 and 05.

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TABLE I. Electrical performance characteristics for device types 04 and 05 - Continued.

Characteristic	Symbol	Conditions 1/		Limits		Unit
				Min	Max	
Response time, high-to-low level collector output	t_{RHLC}	V_{OD} (overdrive) = +5 mV, C_L = 50 pF, V_{IN} = -100 mV	-55°C ≤ T_A ≤ +25°C		300	ns
			T_A = +125°C		500	ns

1/ Unless otherwise specified, V_{IC} = 0 V, $\pm V_{CC}$ = ±15 V and -55°C ≤ T_A ≤ +125°C. Limits apply to each half of device type 05 except as noted for $+I_{CC}$ and $-I_{CC}$.

2/ V_{IC} is achieved by algebraically subtracting the common mode voltage from each V_{CC} (power supplies) and algebraically adding it to V_{IN}. V_{IC} can be calculated by using the following formula:

$$V_{IC} = -\left[\frac{(+V_{CC}) + (-V_{CC})}{2}\right] + V_{IN}$$

3/ Subscript (R) indicates tests which are performed with input stage current raised by connecting BAL and BAL/STB terminals to $+V_{CC}$.

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TABLE I. Electrical performance characteristics for device types 06 and 07.

Characteristics	Symbol	Conditions 1/		Device type	Limits		Unit
					Min	Max	
Input offset voltage	V_{IO}	R_S = 50Ω, V_{IC} = 0 V, +12 V and -12 V 2/	T_A = +25°C	06	-4.0	+4.0	mV
				07	-1.0	+1.0	mV
			T_A = -55°C, +125°C	06	-7.0	+7.0	mV
				07	-2.0	+2.0	mV
		R_S = 50Ω, V_{IC} = 2.5 V, $+V_{CC}$ = 5 V, $-V_{CC}$ = 0 V	T_A = +25°C	06	-4.0	+4.0	mV
				07	-1.0	+1.0	mV
			T_A = -55°C, +125°C	06	-7.0	+7.0	mV
				07	-2.0	+2.0	mV
Input offset current	I_{IO}	V_{IC} = 0 V, +12 V, -12 V 2/	T_A = +25°C, +125°C	06	-75	+75	nA
				07	-40	+40	nA
			T_A = -55°C	06	-100	+100	nA
				07	-75	+75	nA
Input bias current	$+I_{IB}$	V_{IC} = 0 V	T_A = +25°C, +125°C	06, 07	-0.1	+500	nA
			T_A = -55°C	06, 07	-0.1	1000	nA
		V_{IC} = +12 V and -12 V	T_A = +25°C, +125°C	06, 07	-0.1	+750	nA
			T_A = -55°C	06, 07	-0.1	+1000	nA
	$-I_{IB}$	V_{IC} = 0 V	T_A = +25°C, +125°C	06, 07	-0.1	+500	nA
			T_A = -55°C	06, 07	-0.1	+1000	nA
		V_{IC} = +12 V and -12 V	T_A = +25°C, +125°C	06, 07	-0.1	+750	nA
			T_A = -55°C	06, 07	-0.1	+1000	nA
Common mode rejection ratio	CMRR	R_S = 50Ω, V_{IC} = +12 V and -12 V		06, 07	90		dB
Output leakage current	I_{CEX}	$\pm V_{CC}$ = ±18 V, V_O = 18 V	T_A = +25°C	06, 07	-1.0	+2.0	μA
			T_A = +125°C	06, 07	-1.0	+10.0	μA

See footnotes at end of device types 06 and 07.

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TABLE I. Electrical performance characteristics for device types 06 and 07 - Continued.

Characteristics	Symbol	Conditions 1/		Device type	Limits		Unit
					Min	Max	
Positive supply current	$+I_{CC}$		T_A = -55°C	06, 07		+11.5	mA
			T_A = +25°C	06, 07		+10.0	mA
			T_A = +125°C	06, 07		+10.0	mA
Negative supply current	$-I_{CC}$		T_A = -55°C	06, 07	-6.0		mA
			T_A = +25°C	06, 07	-5.0		mA
			T_A = +125°C	06, 07	-4.5		mA
Low level output voltage	V_{OL1}	$+V_{CC}$ = 4.5 V, $-V_{CC}$ = 0 V, V_{ID} = 7.0 mV, I_O = 3.2 mA	V_{IC} = +2.25 V, +1.0 V T_A = -55°C	06, 07		+0.6	V
	V_{OL2}		T_A = +25°C, +125°C	06, 07		+0.4	V
	V_{OL3} V_{OL4}	I_O = 25 mA, V_{ID} = -7.0 mV	V_{IC} = ±12 V	06, 07		+1.5	V
Voltage gain (collector)	A_V		T_A = +25°C	06	+10		V/mV
				07	+20		V/mV
			T_A = -55°C, +125°C	06	+ 5		V/mV
				07	+10		V/mV
Response time, low-to-high level collector output	t_{RLHC}	V_{OD} (overdrive) = -5 mV, V_{IN} = 100 mV, C_L = 50 pF minimum	T_A = -55°C, +25°C	06, 07		125	ns
			T_A = +125°C	06, 07		160	ns
Response time, high-to-low level collector output	t_{RHLC}	V_{OD} (overdrive) = +5 mV, V_{IN} = 100 mV, C_L = 50 pF minimum	T_A = -55°C, +25°C	06, 07		160	ns
			T_A = +125°C	06, 07		160	ns

1/ Unless otherwise specified, V_{IC} = 0 V, $\pm V_{CC}$ = ±15 V and -55°C ≤ T_A ≤ +125°C. Limits apply to each half of device types 06 and 07 except for $+I_{CC}$ and $-I_{CC}$.

2/ V_{IC} is achieved by algebraically subtracting the common mode voltage from each V_{CC} (power supplies) and algebraically adding it to V_{IN}. V_{IC} and can be calculated by using the following formula:

$$V_{IC} = -\left[\frac{(+V_{CC}) + (-V_{CC})}{2}\right] + V_{IN}$$

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Device types	01				02			
Case outlines	C	G	H	2	C	H	I	2
Terminal number	Terminal symbol							
1	NC	GND	GND	NC	NC	INPUT- 1	GND	NC
2	GND	INPUT+	INPUT+	GND	INPUT- 1	INPUT+ 1	STROBE 1	NC
3	INPUT+	INPUT-	INPUT-	NC	INPUT+ 1	V_{CC-}	INPUT- 1	INPUT- 1
4	INPUT-	V_{CC-}	NC	NC	V_{CC-}	INPUT+ 2	INPUT+ 1	INPUT+ 1
5	NC	NC	V_{CC-}	INPUT+	INPUT+ 2	INPUT- 2	V_{CC-}	NC
6	V_{CC-}	NC	OUTPUT	NC	INPUT- 2	STROBE 2	INPUT+ 2	V_{CC-}
7	NC	OUTPUT	NC	INPUT-	NC	OUTPUT	INPUT- 2	NC
8	NC	V_{CC+}	V_{CC+}	NC	NC	V_{CC+}	STROBE 2	INPUT+ 2
9	OUTPUT	---	NC	NC	STROBE 2	GND	OUTPUT	INPUT- 2
10	NC	---	NC	V_{CC-}	OUTPUT	STROBE 1	V_{CC+}	NC
11	V_{CC+}	---	---	NC	V_{CC+}	---	---	NC
12	NC	---	---	NC	GND	---	---	NC
13	NC	---	---	NC	STROBE 1	---	---	STROBE 2
14	NC	---	---	NC	NC	---	---	OUTPUT
15	---	---	---	NC	---	---	---	NC
16	---	---	---	NC	---	---	---	V_{CC+}
17	---	---	---	OUTPUT	---	---	---	NC
18	---	---	---	NC	---	---	---	GND
19	---	---	---	NC	---	---	---	STROBE 1
20	---	---	---	V_{CC+}	---	---	---	NC

NC = No connection

FIGURE 1. Terminal connections.

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Device types	03		04				
Case outlines	A	G	C	G	H	P	2
Terminal number	Terminal symbol						
1	NC	GND	NC	GND	GND	GND	NC
2	GND	INPUT+	GND	INPUT+	INPUT+ 1	INPUT+ 1	GND
3	INPUT+	INPUT-	INPUT+	INPUT-	INPUT- 1	INPUT- 1	NC
4	INPUT-	V_{CC-}	INPUT-	V_{CC-}	NC	V_{CC-}	NC
5	NC	STROBE 1	NC	BAL	V_{CC-}	BAL	INPUT+
6	V_{CC-}	STROBE 2	V_{CC-}	BAL/STB	BAL	BAL/STB	NC
7	STROBE 1	OUTPUT	BAL	OUTPUT	BAL/STB	OUTPUT	INPUT-
8	STROBE 2	V_{CC+}	BAL/STB	V_{CC+}	NC	V_{CC+}	NC
9	OUTPUT	---	OUTPUT	---	OUTPUT	---	NC
10	NC	---	NC	---	V_{CC+}	---	V_{CC-}
11	V_{CC+}	---	V_{CC+}	---	---	---	NC
12	NC	---	NC	---	---	---	BAL
13	NC	---	NC	---	---	---	NC
14	NC	---	NC	---	---	---	NC
15	---	---	---	---	---	---	BAL/STB
16	---	---	---	---	---	---	NC
17	---	---	---	---	---	---	OUTPUT
18	---	---	---	---	---	---	NC
19	---	---	---	---	---	---	NC
20	---	---	---	---	---	---	V_{CC+}

NC = No connection
BAL = Balance
BAL/STB = Balance/Strobe

FIGURE 1. Terminal connections - Continued.

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impact on pins 2 and the third impact on pins 7. The first impact shall

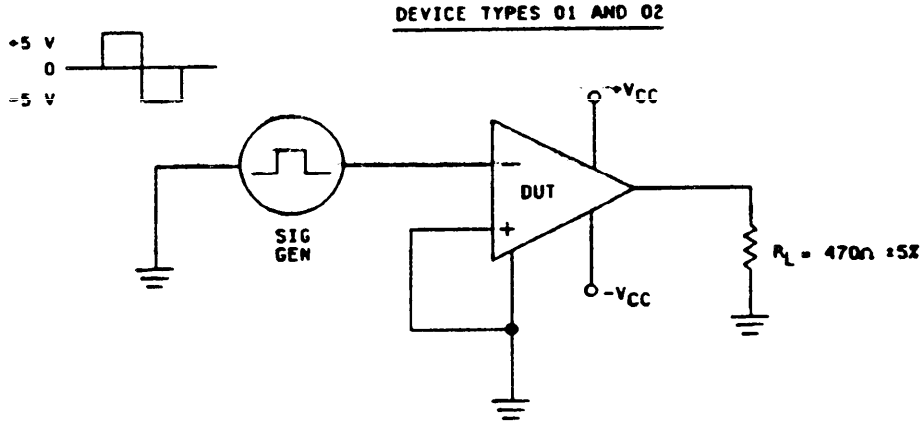
Device types	05	06 and 07		
Case outlines	E and F	C	H	I
Terminal number	Terminal symbol			
1	V_{CC+} A	NC	OUTPUT 1	OUTPUT 1
2	GND A	NC	GND 1	GND 1
3	INPUT+ A	GND 1	INPUT+ 1	INPUT+ 1
4	INPUT- A	INPUT+ 1	INPUT- 1	INPUT- 1
5	V_{CC-}	INPUT- 1	V_{CC-}	V_{CC-}
6	BAL B	V_{CC-}	OUTPUT 2	OUTPUT 2
7	BAL/STB B	OUTPUT 2	GND 2	GND 2
8	COLLECTOR OUTPUT B	GND 2	INPUT+ 2	INPUT+ 2
9	V_{CC+} B	INPUT+ 2	INPUT- 2	INPUT- 2
10	GND B	INPUT- 2	V_{CC+}	V_{CC+}
11	INPUT+ B	V_{CC+}	---	---
12	INPUT- B	OUTPUT 1	---	---
13	BAL A	NC	---	---
14	BAL/STB A	NC	---	---
15	COLLECTOR OUTPUT A	---	---	---
16	NC	---	---	---
17	---	---	---	---
18	---	---	---	---
19	---	---	---	---
20	---	---	---	---

NC = No connection
BAL = Balance
BAL/STB = Balance/Strobe

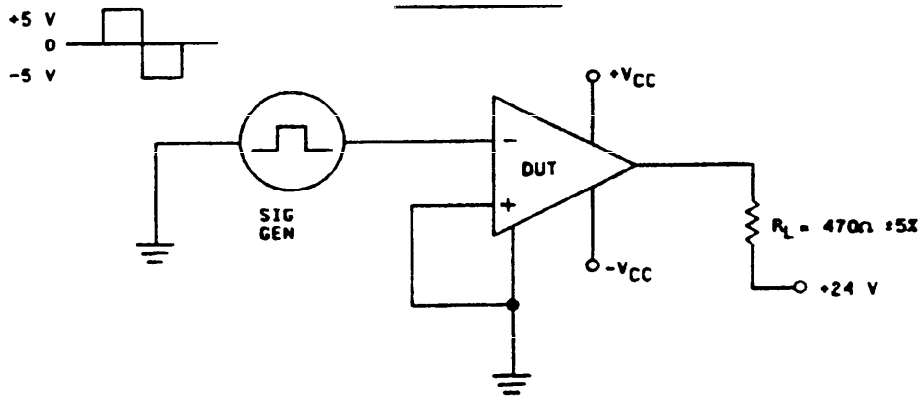
FIGURE 1. Terminal connections - Continued.

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DEVICE TYPES 01 AND 02



DEVICE TYPE 03

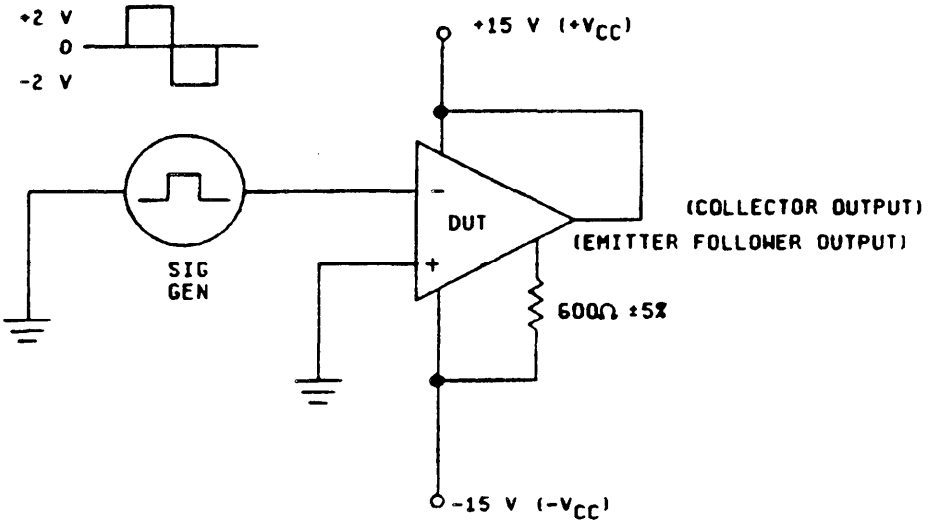


NOTES:
1. Test conditions: T_A = +125°C; $+V_{CC}$ = 12 V; $-V_{CC}$ = -6 V; signal generator output = 10 V(p-p), f = 5 Hz minimum.
2. Input applied to both channels of device type 02, strobe inputs open.
3. For device type 03, strobe inputs open.

FIGURE 2. Test circuit, burn-in and steady-state life test.

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DEVICE TYPES 04 AND 05

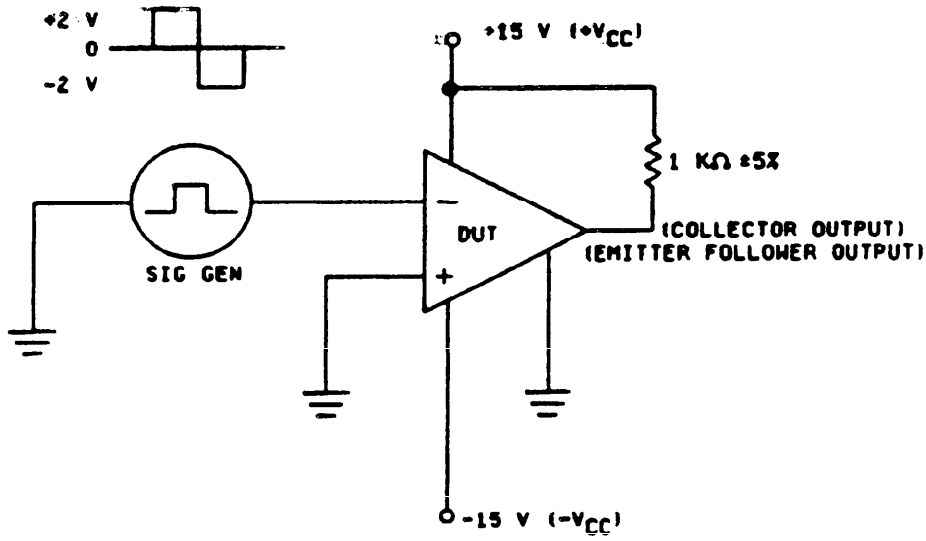


NOTES:
1. Signal generator output is 4 V(p-p), 50 percent duty cycle at 1 kHz.
2. Strobe and balance inputs are open.
3. T_A = +125°C.
4. For device type 05, both halves shall be connected as above.

FIGURE 2. Test circuit, burn-in and steady-state life test - Continued.

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DEVICE TYPES 06 AND 07

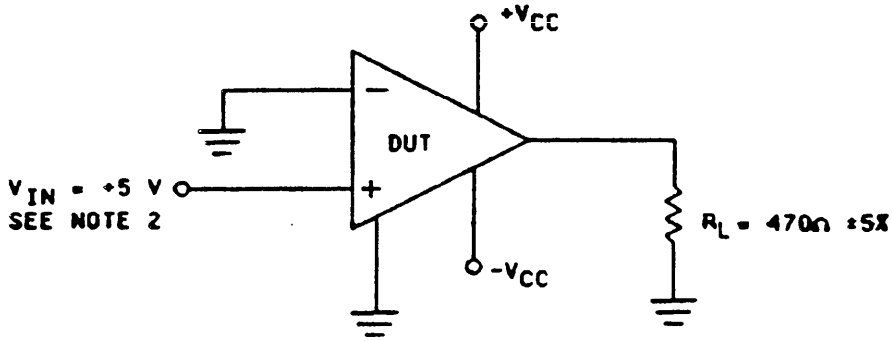


NOTES:
1. Signal generator output is 4 V(p-p), 50 percent duty cycle at 1 kHz.
2. T_A = +125°C.
3. Both halves shall be connected as above.

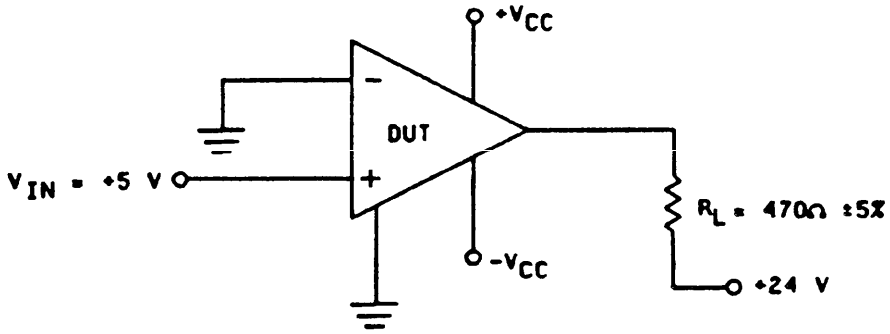
FIGURE 2. Test circuit, burn-in and steady-state life test - Continued.

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DEVICE TYPES 01 AND 02



DEVICE TYPE 03

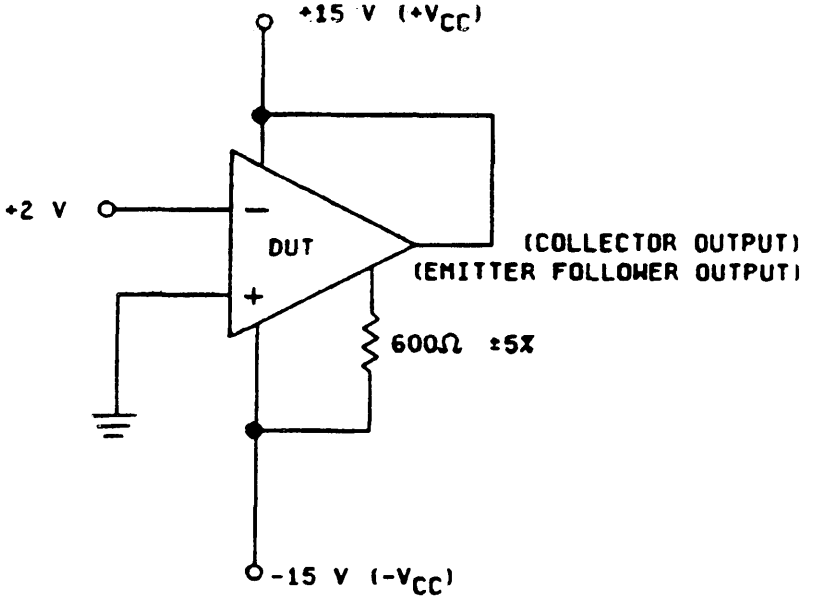


NOTES:
1. Test conditions: T_A = +125°C; $+V_{CC}$ = 12 V; $-V_{CC}$ = -6 V.
2. For device type 02, V_{IN} shall be positive into both channels.
3. For device type 03, strobe inputs open.

FIGURE 3. Test circuit, burn-in (steady-state power and reverse bias) and steady-state life test.

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DEVICE TYPES 04 AND 05

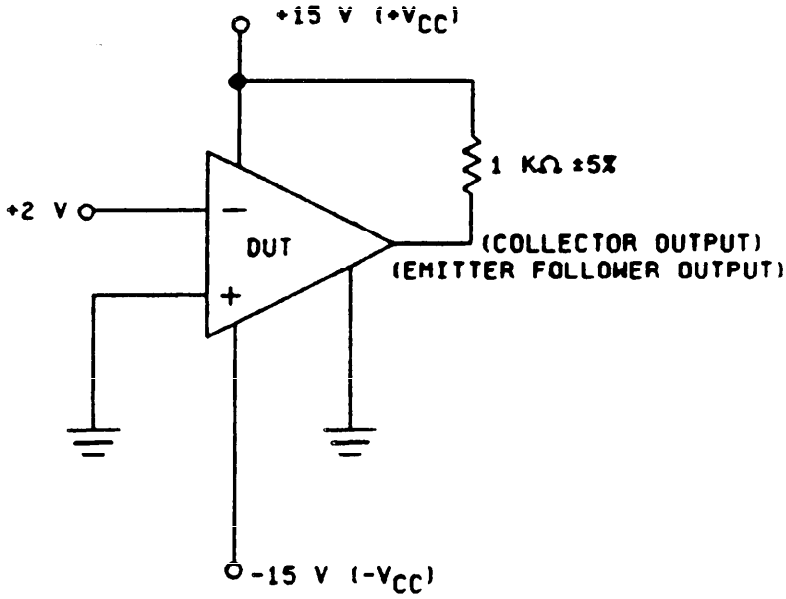


NOTES:
1. Strobe and balance inputs are open.
2. T_A = +125°C.
3. For device type 05, both halves shall be connected as above.

FIGURE 3. Test circuit, burn-in (steady-state power and reverse bias) and steady-state life test - Continued.

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DEVICE TYPES 06 AND 07

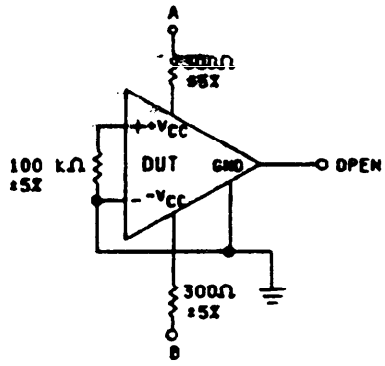


NOTES:
1. T_A = +125°C.
2. For device types 06 and 07, both halves shall be connected as above.

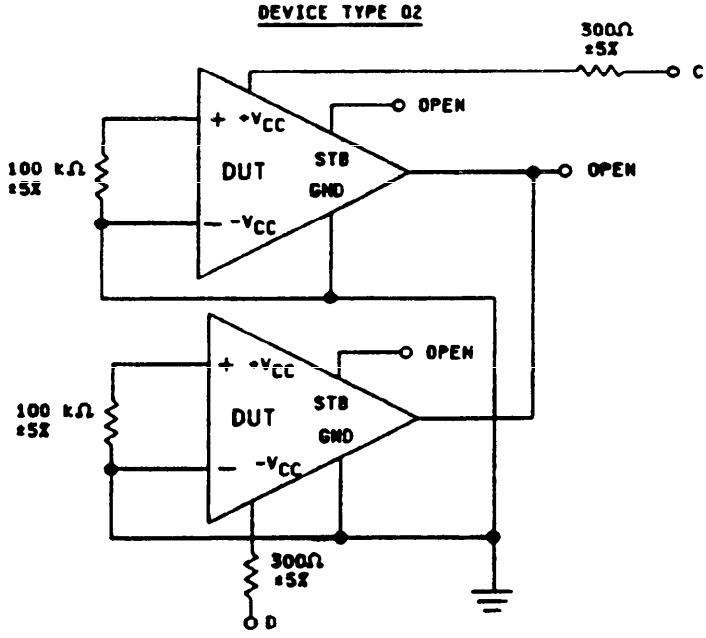
FIGURE 3. Test circuit, burn-in (steady-state power and reverse bias) and steady-state life test - Continued.

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DEVICE TYPES 01 AND 03



DEVICE TYPE 02

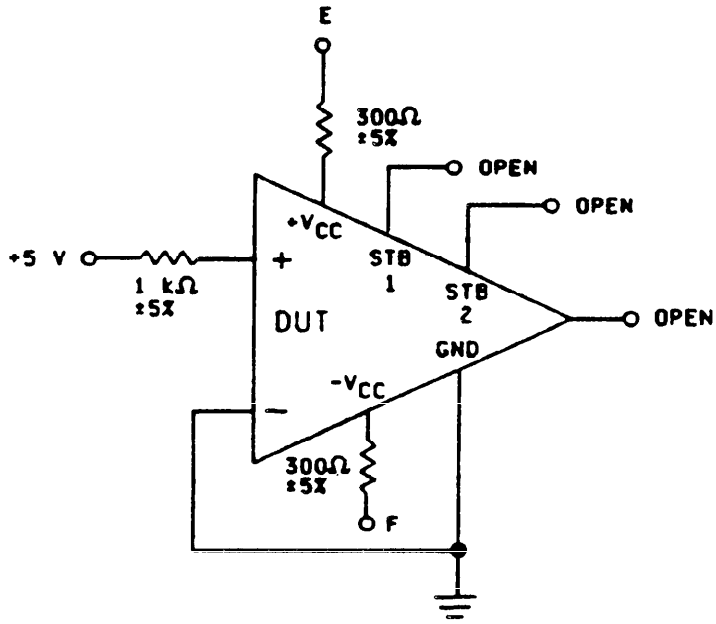


NOTE: For device types 01 and 03, the value of A and B shall be sufficient to maintain +12 V at $+V_{CC}$ and -6 V at $-V_{CC}$. For device type 02, the value of C and D shall be sufficient to maintain +14 V at $+V_{CC}$ and -7 V at $-V_{CC}$.

FIGURE 4. Accelerated burn-in and life test circuits.

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DEVICE TYPES 04 AND 05

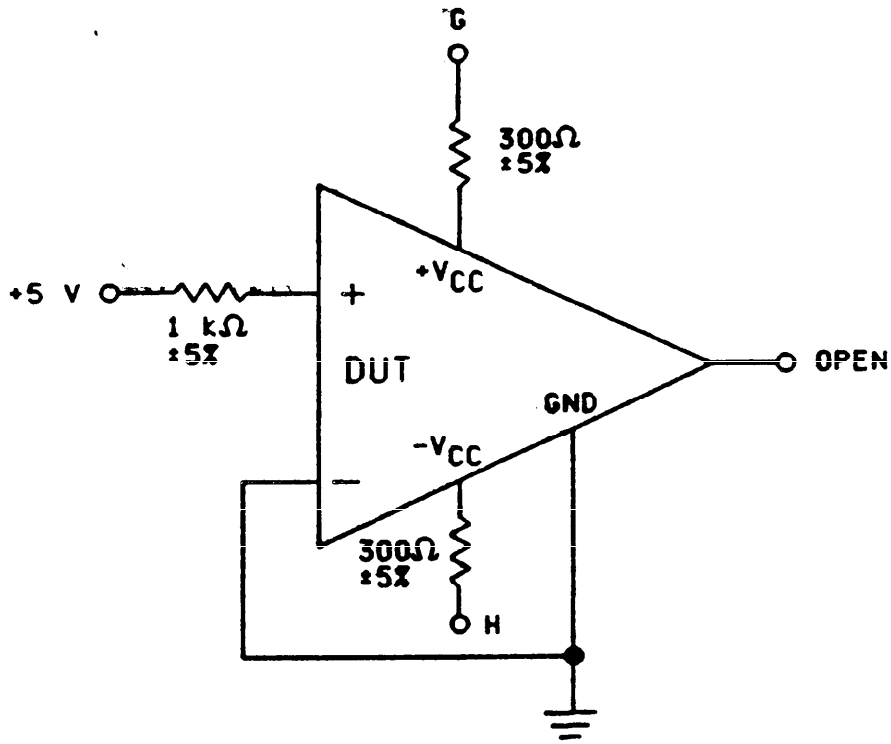


NOTES:
1. The value of E and F shall be sufficient to maintain +15 V at +V_{CC} and -15 V at -V_{CC}.
2. Both halves of device type 05 shall be connected as shown.

FIGURE 4. Accelerated burn-in and life test circuits - Continued.

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DEVICE TYPES 06 AND 07



NOTES:
1. The value of G and H shall be sufficient to maintain +15 V at $+V_{CC}$ and -15 V at $-V_{CC}$.
2. Both halves shall be connected as shown.

FIGURE 4. Accelerated burn-in and life test circuits - Continued.

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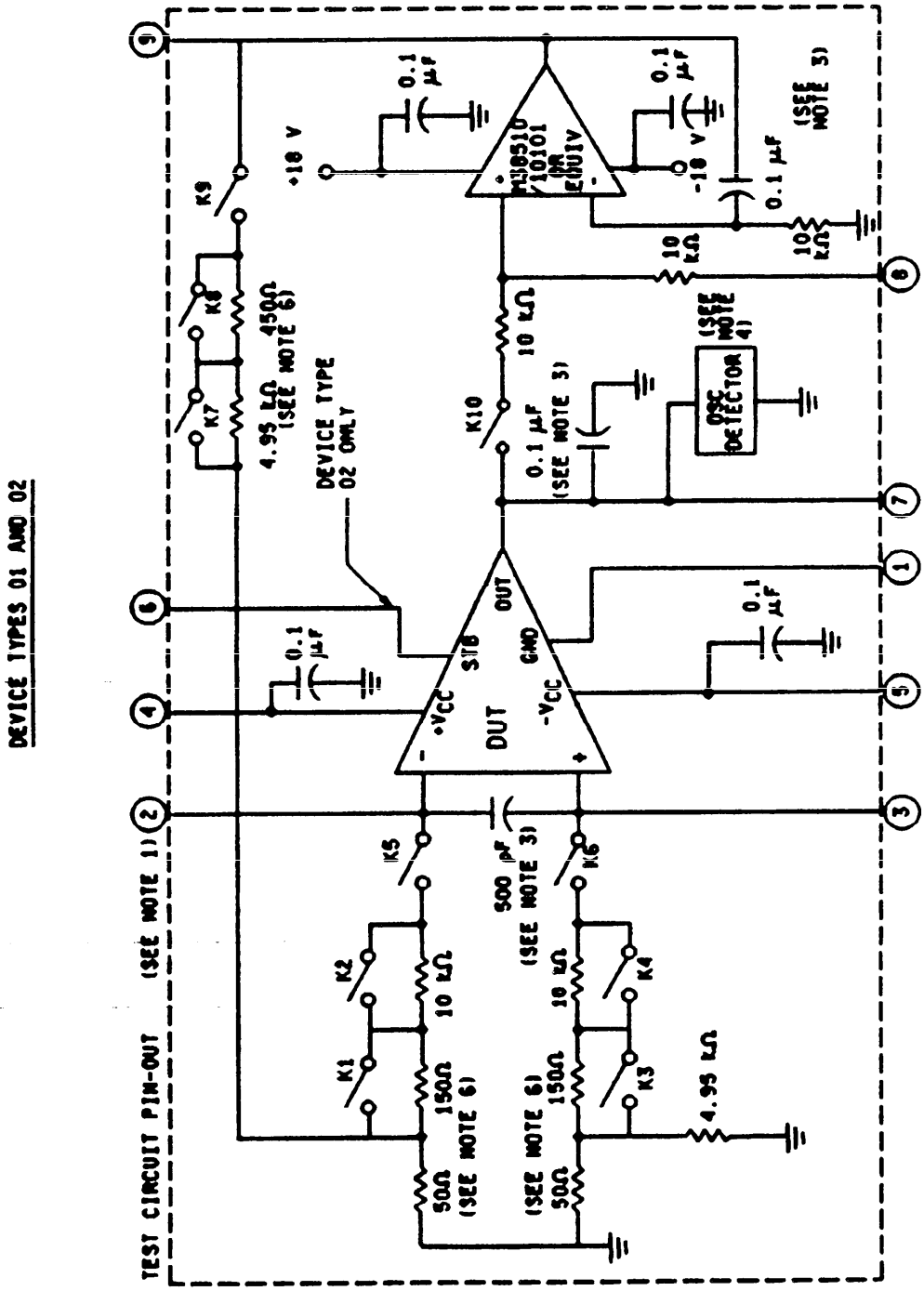


FIGURE 5. Test circuit for static and dynamic tests.

See notes at end of figure.

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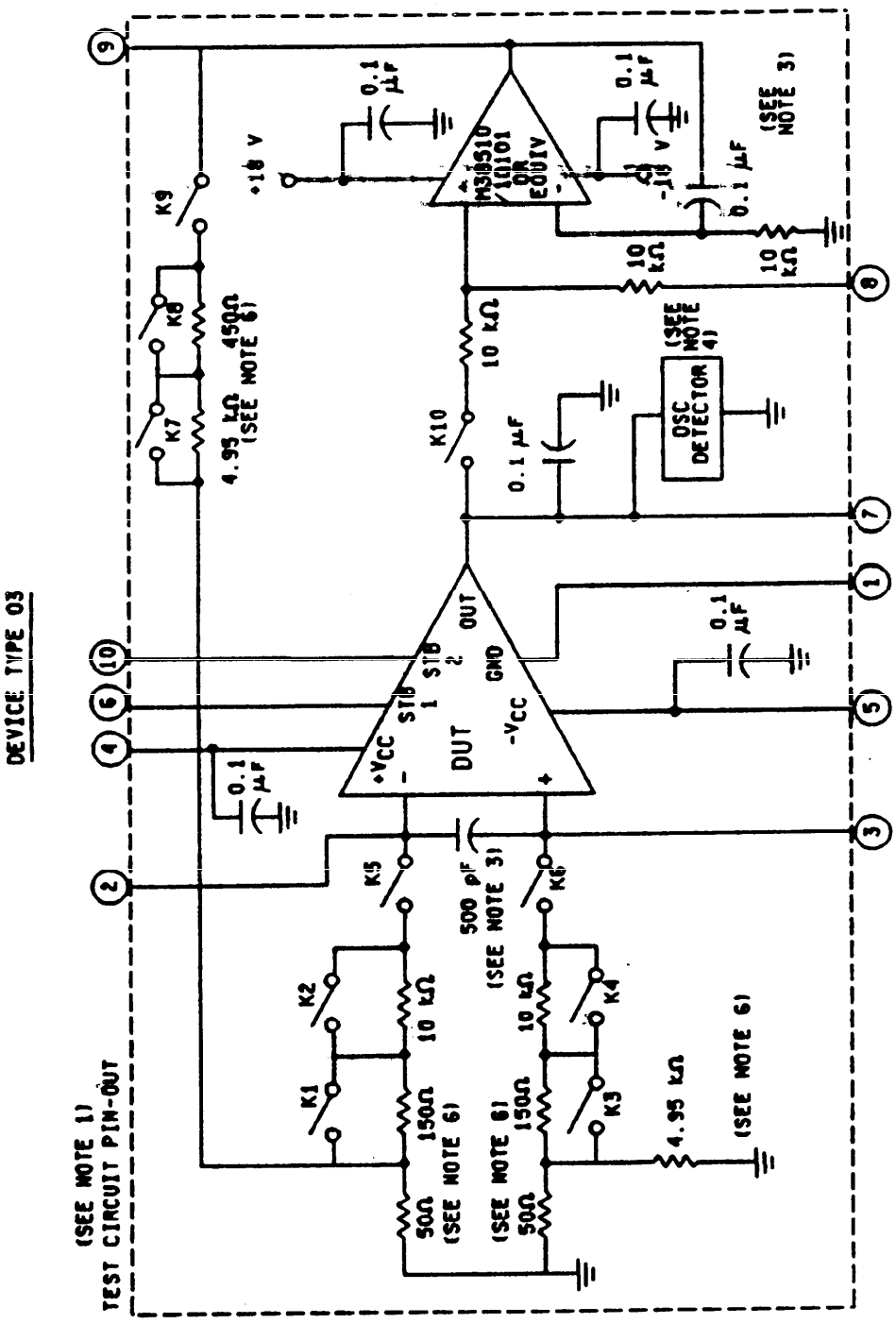


FIGURE 5. Test circuit for static and dynamic tests - Continued.

See notes at end of figure.

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DEVICE TYPES 04 AND 05

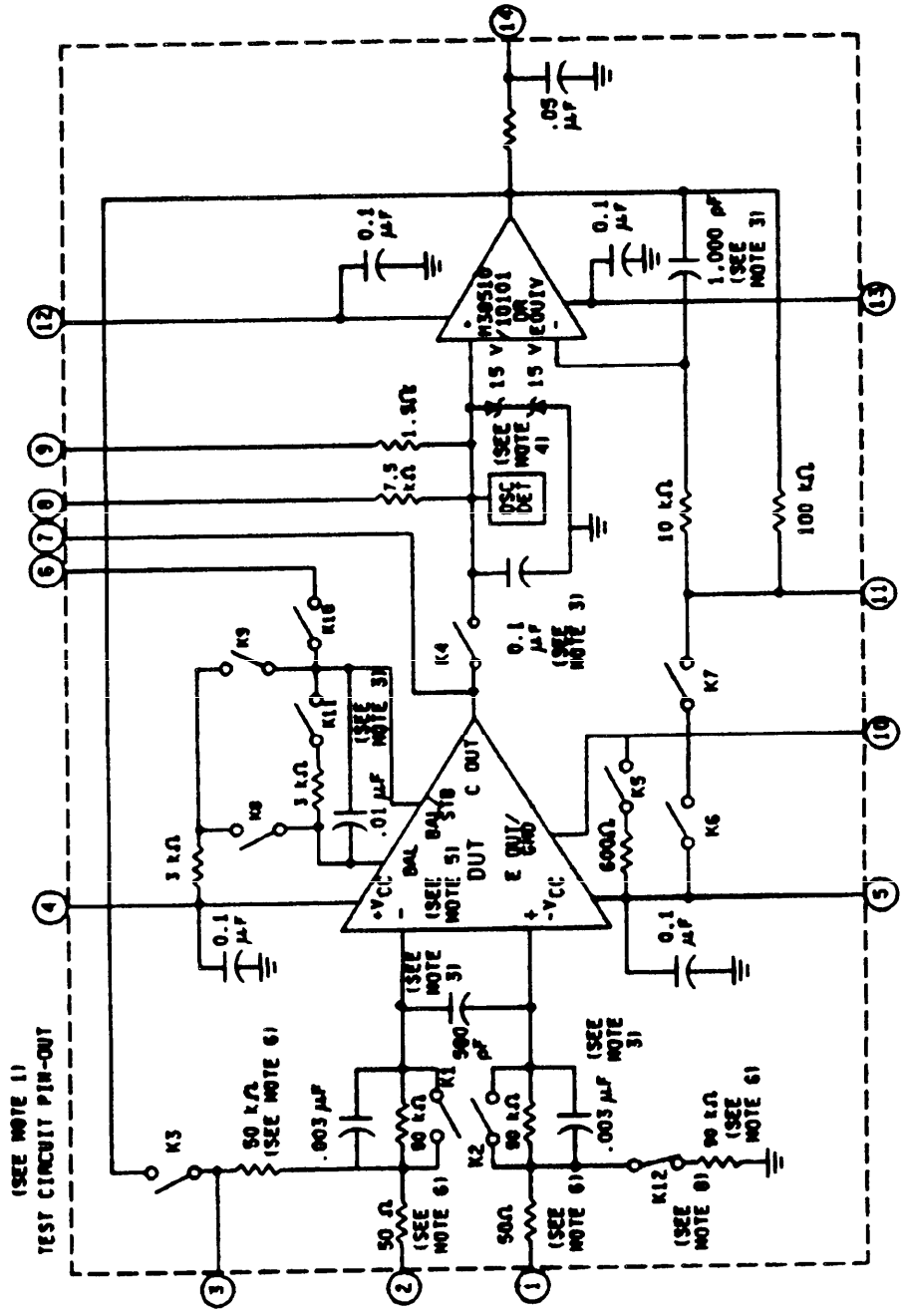


FIGURE 5. Test circuit for static and dynamic tests - Continued.

See notes at end of figure.

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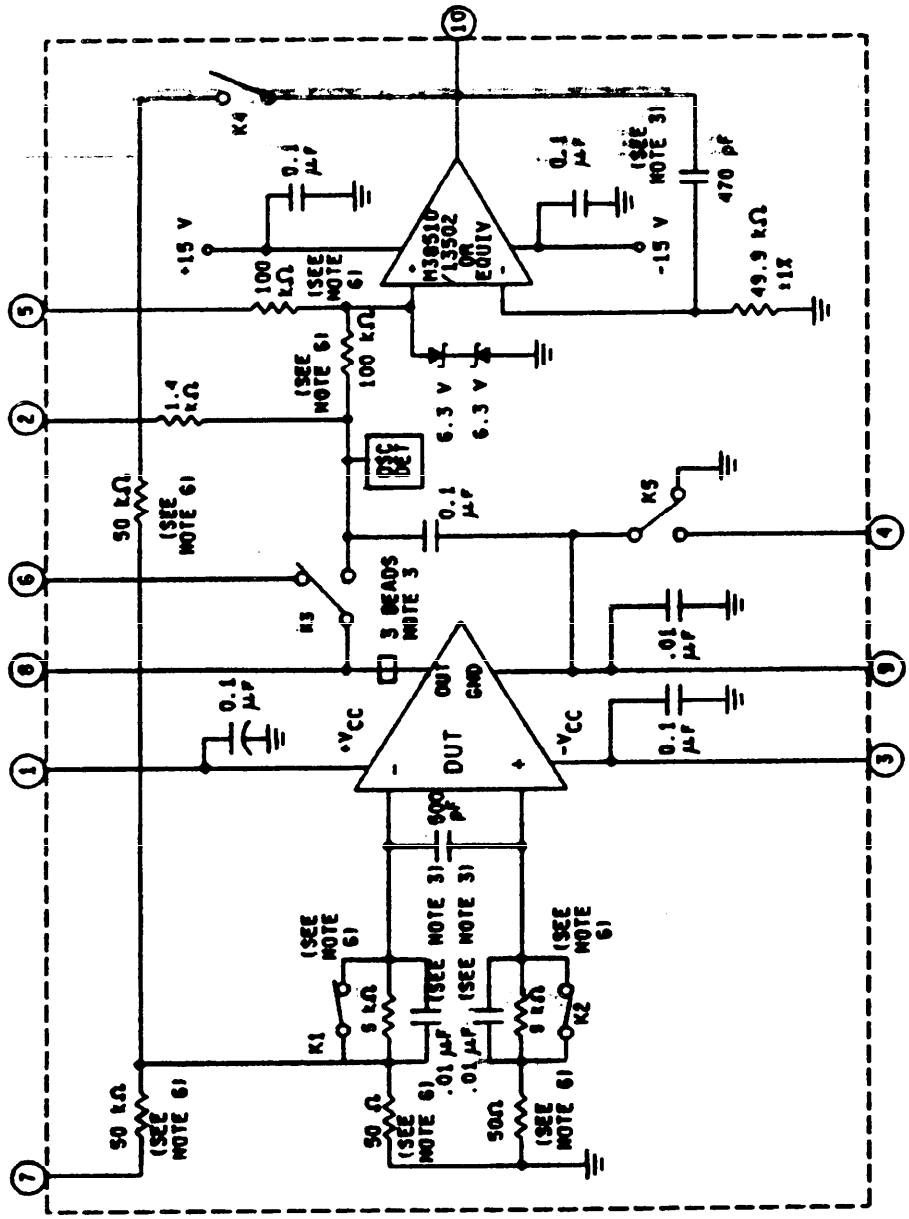


FIGURE 5. Test circuit for static and dynamic tests - Continued.

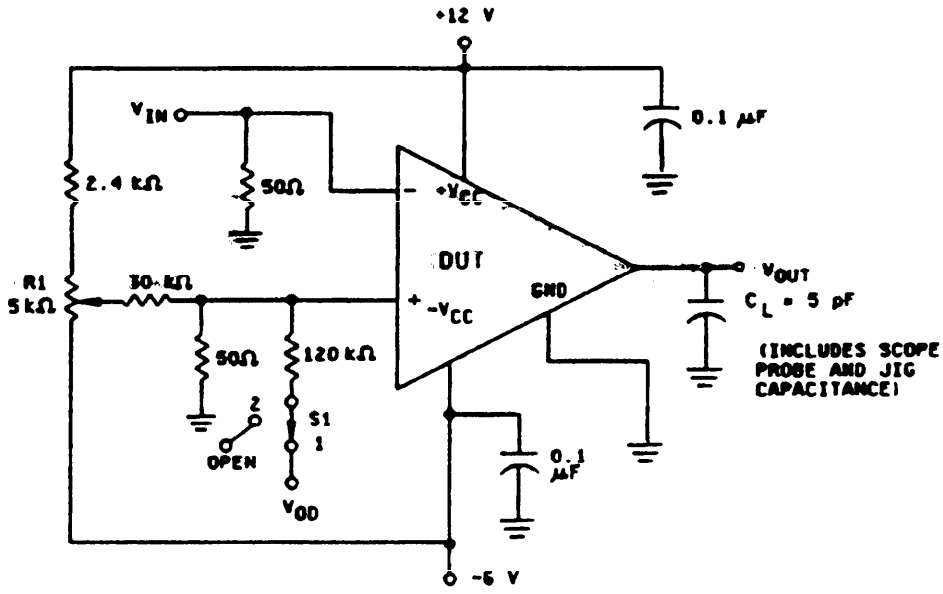
See notes at end of figure.

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NOTES:
1. Test circuit pin conditions and test temperatures shall be as specified in table III.
2. Precautions shall be taken to prevent damage to the DUT during insertion into socket and change of relay switch positions (e.g., disable voltage-current supplies, current limit $\pm V_{CC}$, etc.).
3. As required to prevent oscillations. Also, proper wiring procedures shall be followed to prevent oscillations. Loop response and settling time shall be consistent with test rate such that any value has settled to within 5 percent if its final value before value is measured. Suggested values shown may not ensure loop stability for all layouts. Actual compensation used shall be approved by the preparing activity prior to use.
4. Any oscillation greater than 300 mV(p-p) shall be cause for device failure.
5. For dual devices, both halves shall be tested. The output of the idle half shall be driven to the off state by using either a differential input voltage or a stobe input, or the idle half may be biased the same as the tested half if oscillations can be avoided.
6. These resistors are ±0.1 percent tolerance matched to ±0.01 percent. All other resistors are ±1 percent tolerance and capacitors are 10 percent tolerance.
7. All relays are shown in the normal deenergized state.

FIGURE 5. Test circuit for static and dynamic tests - Continued.

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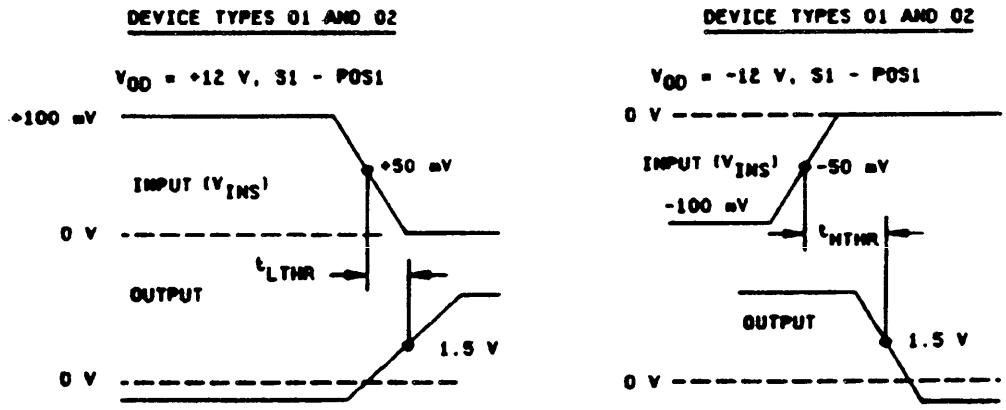
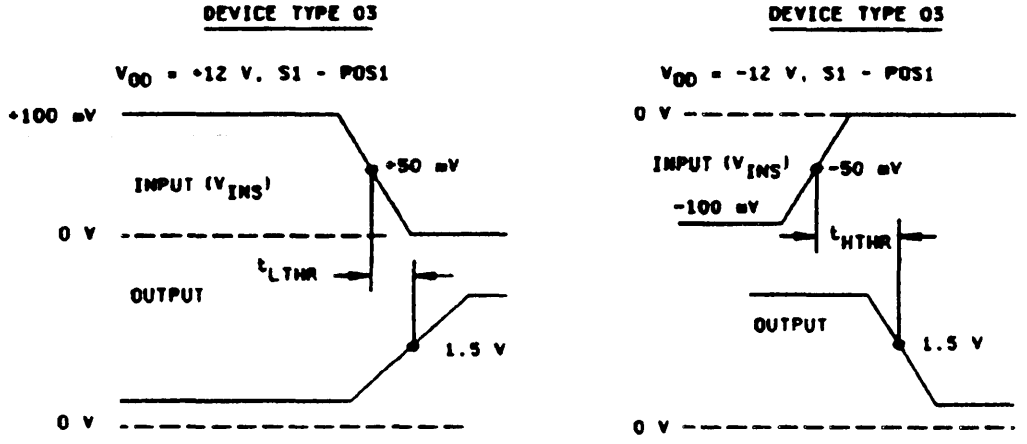


FIGURE 6. Response time test circuit and waveforms for device types 01, 02, and 03.

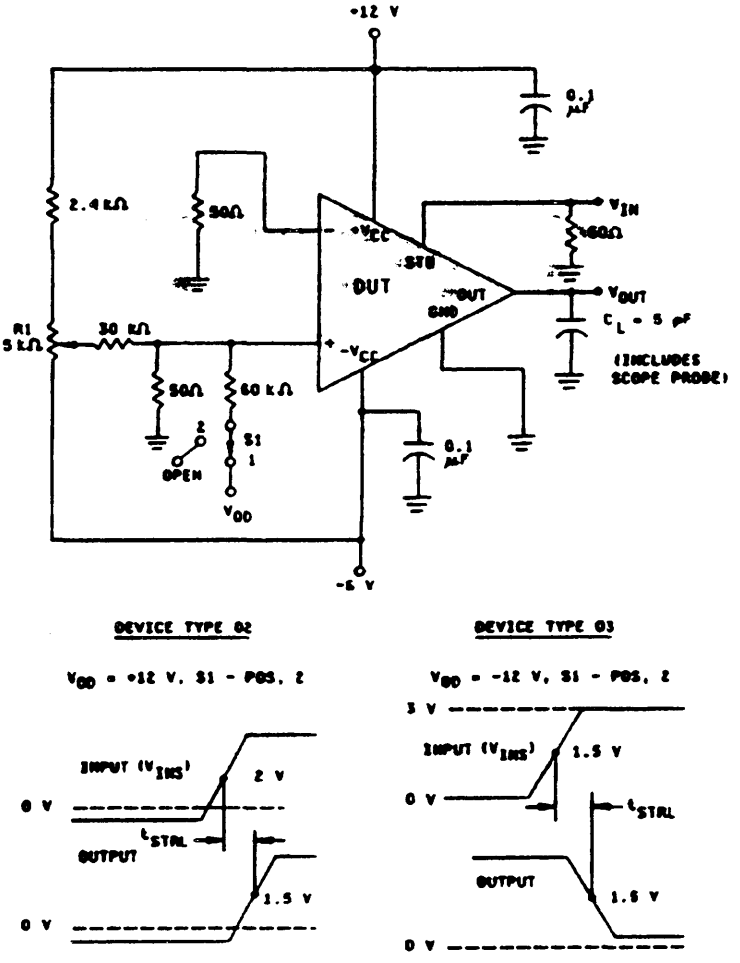
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NOTES:
1. V_{IN} = 100 ns pulse width, 100 kHz repetition rate, t_r and $t_f \leq 5$ ns (see waveforms).
2. Setup procedure: With S1 open and V_{IN} = 0 adjust R1 for V_{OUT} = 1.5 V. Apply V_{DD} (see note 5) and close S1, then apply V_{IN}. (R1 voltage divider may be replaced with a variable power supply.)
3. All resistor tolerances are ±1 percent and all capacitor tolerances are ±10 percent.
4. For device type 02, ground strobe on idle side. For device type 03, strobe is left open.
5. See waveforms for V_{DD} values.
6. When V_{DD} = 12 V, overdrive applied to DUT = 5 mV.

FIGURE 6. Response time test circuit and waveforms for device types 01, 02, and 03 - Continued.

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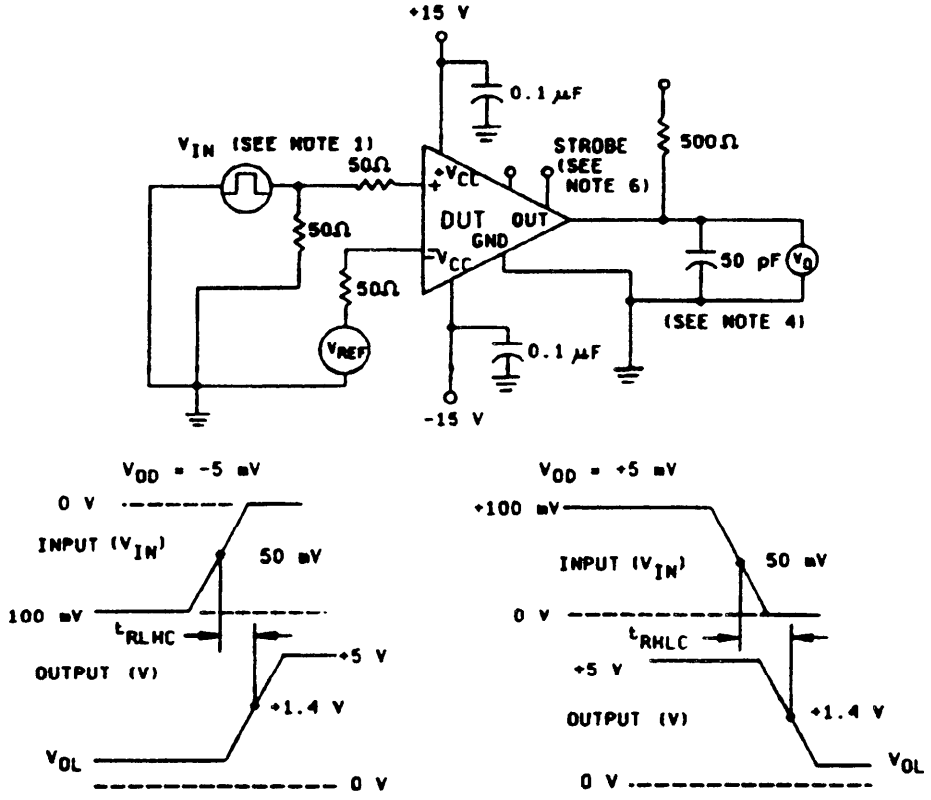


NOTES:

1. V_{IN} = 100 ns pulse width, 100 kHz repetition rate, t_r and $t_f \leq 5$ ns (see waveforms).
2. Setup procedure: With S1 open, V_{IN} = 3 V (device type 02), and V_{IN} = 0 V (device type 03) adjust R1 for V_{OUT} = 1.5 V. Apply V_{OD} (see note 5) and close S1, then apply V_{IN}. (R1 voltage divider may be replaced with a variable power supply.)
3. All resistor tolerances are ±1 percent and all capacitor tolerances are ±10 percent.
4. For device type 02, ground strobe on idle side. For device type 03, unused strobe is left open.
5. See waveforms for V_{OD} values.
6. When V_{OD} = 12 V, overdrive applied to DUT = 10 mV.

FIGURE 7. Strobe response time test circuit and waveforms for device types 02 and 03.

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NOTES:

1. V_{IN} = 10 µs pulse width at 50 kHz, t_{TLH} and $t_{THL} \leq$ 10 ns, and Z_O = 50Ω.
2. Setup procedure:
   a. With V_{IN} = 0 V, step V_{REF} from -3.0 mV to +3.0 mV in 0.1 mV steps for subgroup 7, -4.0 mV to +4.0 mV for subgroup 8, and stop when output switches from high to low.
   b. Change V_{REF} from the reference value obtained above by the required V_{OD} (overdrive).
   c. Apply V_{IN} and measure response time.
3. All resistor tolerances are ±1 percent and all capacitor tolerances are ±10 percent.
4. C_L includes scope, probe, and jig capacitance.
5. T_A = +25°C for subgroup 7. For subgroup 8, the tests shall be performed twice, at T_A = +125°C and T_A = -55°C.
6. In an environment with excessive noise, a 0.01 µF capacitor may be placed between balance and balance/strobe pins.

FIGURE 8. Response time test circuit and waveforms for collector output for device types 04 and 05.

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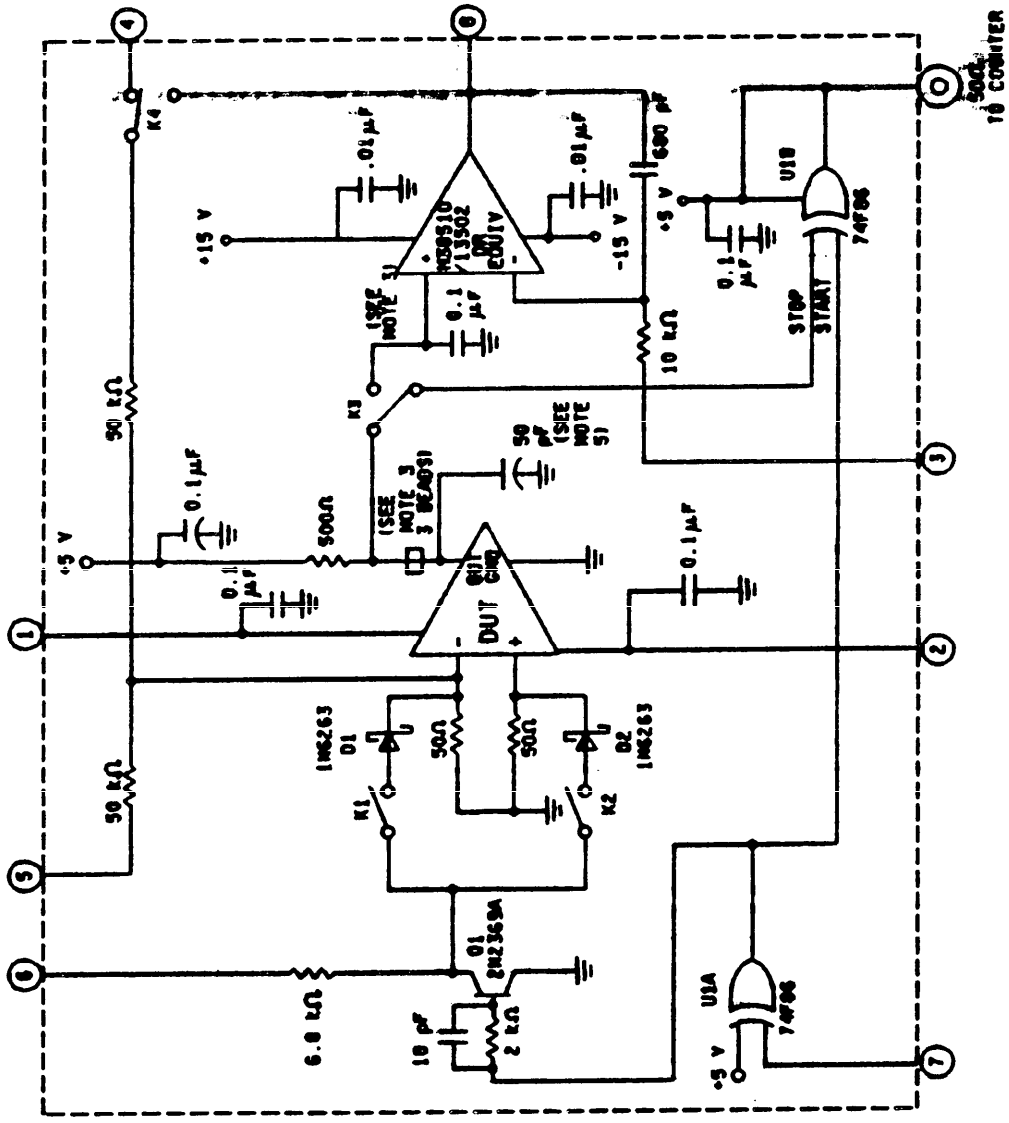


FIGURE 9. Response time test circuit for device types 06 and 07.

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NOTES:

1. This circuit is designed especially to be used with a computer-controlled automatic tester, although it can also be implemented as a bench test setup. The test table for subgroups 9, 10, and 11 lists in detail the steps in a typical test sequence, which goes as follows:
   a. Measure V_{IO}: DUT is in a conventional servo loop (K3 and K4 energized) with output served to TTL logic threshold. (1.4 V on adapter pin 3). Measure V_{IO} x 1000 at adapter pin 8.
   b. Null V_{IO}: Release K3 and K4, apply voltage measured in step 1 to adapter pin 4. This is particularly easy to do on computer-controlled automatic test equipment.
   c. Apply 5 mV overdrive (OD): The overdrive is developed at the inverting input of the DUT via a 1000:1 divider from adapter pin 5.
   d. Apply initial 100 mV: Close K1, apply 2 mA at adapter pin 6. Since Q1 is off, this current flows through Schottky diode D1 and the 50Ω source resistor, giving 100 mV. If a current source is not available, 15 V at pin 6 will give about 2 mA current.
   e. Measure t_{RLHC}: Pull adapter pin 7 low. U1A output goes high, starting the timer via U1B and turning on Q1, which turns off D1. The 100 mV drive disappears abruptly, leaving just the 5 mV overdrive. After the response time, the DUT output crosses TTL logic threshold and stops the timer via U1B. On a bench setup, the pulse at U1B output can be measured with a scope.
   f. Measure t_{RHLC}: Reset pin 7, change pin 5 from -5 V to +5 V, open K1, close K2, pull pin 7 low, measure pulse width at U1B output.

2. The following delay times are of possible concern in the LM119 (device types 06 and 07) response time test circuit:
   a. U1A: This device merely inverts and squares up the start signal from the tester. Its gate delay does not affect the tests.
   b. Q1: This is the major uncompensated delay, since the timer starts when Q1 base goes high, not when Q1 collector goes low. This delay is minimized by using a fast switching transistor with an R-C speedup network driving the base. Measured Q1 delays are well under 10 ns.
   c. D1 and D2: These Schottky devices have negligible switching times (< 1 ns).
   d. U1B: The gate delays here are not important as long as the delays from the two inputs are well matched. (Matched delays merely offset the output pulse in time.) To minimize gate delay effects even further, we use one of the "fast" series 74F86 gates; with delays around 5 ns from each input, the mismatch should be no more than 1 or 2 ns.
   e. Q2: This driver is required only if the time measurement system has a 50Ω input. Since it operates as an emitter follower rather than a saturating switch, there are no delays associated with it.

3. As required to prevent oscillations. Also, proper wiring procedures shall be followed to prevent oscillations. Loop response and settling time shall be consistent with test rate such that any value has settled to within 5 percent of its final value before value is measured. Suggested values shown may not ensure loop stability for all layouts. Actual compensation used shall be approved by preparing activity prior to use.

4. All resistor tolerances are ±1 percent and all capacitor tolerances are ±10 percent.

5. C_L includes scope, probe, and jig capacitance.

FIGURE 9. Response time test circuit for device types 06 and 07 - Continued.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	2/ 3/ Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004) 1/	1, 2, 3, 4	1, 2, 3, 4
Group A test requirements (method 5005)	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11	1, 2, 3, 4, 5, 6, 7, 8, 9
Group C end points and group B, class S, electrical parameters (method 5005)	1, 2, 3, and table IV delta limits	1 and table IV delta limits
Group D end-point electrical parameters (method 5005)	1, 2, 3	1

1/ PDA applies to subgroup 1 (see 4.2d).
2/ Subgroup 8 applies to device types 04 and 05 only.
3/ Subgroups 9, 10, and 11 apply to device types 06 and 07.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) For class S devices: Test condition D using the circuit shown on figure 2.

(2) For class B devices: Test condition D using the circuit shown on figure 2, test condition C using the circuit shown on figure 3 or test condition F using the circuit shown on figure 4.

b. Reverse bias burn-in (method 1015 of MIL-STD-883). For class S devices only using the circuit shown on figure 3.

c. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

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d. The percent defective allowable (PDA) for classes S and B devices shall be as specified in MIL-M-38510 based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1, 4.4.2, and 4.4.3).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883, and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883. Tests shall be as specified in table II herein.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table IV herein.

b. Steady-state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using the circuit shown on figure 2. If the alternate burn-in conditions are used, the circuit shown on figure 3 shall be used.

4.4.3 Groups C and D inspections. Groups C and D inspections shall be in accordance with tables III and IV of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Life test for class B (method 1005 of MIL-STD-883). Test condition D using the circuit shown on figure 2, test condition C using the circuit shown on figure 3, or test condition F using the circuit shown on figure 4 (see 4.2a(2)).

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables. Electrical test circuits as prescribed herein or in the referenced test methods of MIL-STD-883 shall be acceptable. Other test circuits shall require the approval of the qualifying activity.

4.5.1 Voltage and current. All voltage values given, except the input offset voltage (or differential voltage) are referenced to the external zero reference level of the supply voltage. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cooldown procedure. When devices are measured at +25°C following application of the steady-state life or burn-in test condition, they shall be cooled to within +10°C of their power stable condition at room temperature prior to removal of the bias.

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TABLE III. Group A inspection for device types 01 and 02.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers 1	2	3	4	5	6	7	8	9	Relays closed
1 T_A = +25°C (figure 5)	V_{IO1}	4001	1	GND			+12.0 V	-6.0 V			-1.4 V		1,2,3,4,5,6,8,9,10
	V_{IO2}	"	2	"			"	"			"		2,4,5,6,8,9,10
	I_{IO}	"	3	"			"	"			"		1,3,5,6,8,9,10
	$+I_{IB}$	"	4	"			"	"			"		1,2,3,5,6,7,9,10
	$-I_{IB}$	"	5	"			"	"			"		1,3,4,5,6,7,9,10
	CMR	4003	6	+5.0 V			+17.0 V	-2.0 V			-6.4 V		2,4,5,6,8,9,10
	CMR	4003	7	-5.0 V			+7.0 V	-12.0 V			+3.6 V		2,4,5,6,8,9,10
	$+I_{CC}$	3005	8	GND		1/	+12.0 V	-6.0 V					1,2,5
	$-I_{CC}$	3005	9	"		"	"	"					1,2,5
	V_{OL}	3007	10	"		"	"	"					1,2,5
	I_{OL}	3009	11	"		"	"	"					1,2,5
	V_{OH1}	3006	12	"	1/		"	"		-5 mA			3,4,6
	V_{OH2}	3006	13	"	"		"	"					3,4,6
	$V_{O(STB)}$ 2/	3007	14	"	"		"	"	300 mV				3,4,6
	$I_{O(STB)}$ 2/	3009	15	"	"		"	"	100 mV				3,4,6
2 T_A = +125°C (figure 5)	V_{IO1}	4001	16	"			"	"			-1.0 V		1,2,3,4,5,6,8,9,10
	V_{IO2}	4001	17	"			"	"			-1.0 V		2,4,5,6,8,9,10
	$\Delta V_{IO}/\Delta T$ 3/		18	Calculation: $\Delta V_{IO}/\Delta T = \left[V_{IO}\text{ (test 16)} - V_{IO}\text{ (test 1)}\right] \times 10^3/100°C$									
	I_{IO}	4001	19	GND			+12.0 V	-6.0 V			-1.0 V		1,3,5,6,8,9,10
	$\Delta I_{IO}/\Delta T$ 3/		20	Calculation: $\Delta I_{IO}/\Delta T = \left[I_{IO}\text{ (test 19)} - I_{IO}\text{ (test 3)}\right] \times 10^3/100°C$									

See footnotes at end of device types 01 and 02.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Pin measured			Equation	Limits					
								Device type 01			Device type 02		
				No.	Value	Unit		Min	Max	Unit	Min	Max	Unit
1 T_A = +25°C (figure 5)	V_{IO1}	4001	1	9	E_1	V	$V_{IO} = 10E_1$	-2	+2	mV	-3.5	+3.5	mV
	V_{IO2}	"	2	"	E_2	"	$V_{IO} = 10E_2$	-2	+2	mV	-3.5	+3.5	mV
	I_{IO}	"	3	"	E_3	"	$I_{IO} = E_1 - E_3$	-3	+3	µA	-10	+10	µA
	$+I_{IB}$	"	4	"	E_4	"	$+I_{IB} = E_1 - 10E_4$	-.1	+20	µA	-.1	+75	"
	$-I_{IB}$	"	5	"	E_5	"	$-I_{IB} = 10E_5 - E_1$		+20	µA	-.1	+75	"
	CMR	4003	6	9	E_6	"	$CMR = 20 \log \left[\frac{10^3}{E_6 - E_7} \right]$	+80		dB	+80		dB
	CMR	4003	7	9	E_7	"							
	$+I_{CC}$	3005	8	4	$+I_{CC}$	mA		+0.5	+9.0	mA	+0.5	+13.5	mA
	$-I_{CC}$	3005	9	5	$-I_{CC}$	mA		-7.0	-0.5	mA	-6.2	-0.5	mA
	V_{OL}	3007	10	7	V_{OL}	V		-1.0		V	-1.0		V
	I_{OL}	3009	11	"	I_{OL}	mA		+2.0		mA	+0.5		mA
	V_{OH1}	3006	12	"	V_{OH1}	V		+2.5	+5.0	V	+2.5	+5.0	V
	V_{OH2}	3006	13	"	V_{OH2}	V		+2.5	+5.0	V	+2.5	+5.0	"
	$V_{O(STB)}$ 2/	3007	14	"	$V_{O(STB)}$	V					-1.0		"
	$I_{O(STB)}$ 2/	3009	15	6	$I_{O(STB)}$	mA					-2.5	-0.1	mA
2 T_A = +125°C (figure 5)	V_{IO1}	4001	16	9	E_8	V	$V_{IO} = 10E_8$	-3	+3	mV	-4.5	+4.5	mV
	V_{IO2}	4001	17	9	E_9	V	$V_{IO} = 10E_9$	-3	+3	mV	-4.5	+4.5	mV
	$\Delta V_{IO}/\Delta T$ 3/		18					-10	+10	µV/°C	-10	+10	µV/°C
	I_{IO}	4001	19	9	E_{10}	V	$I_{IO} = E_8 - E_{10}$	-3	+3	µA	-10	+10	µA
	$\Delta I_{IO}/\Delta T$ 3/		20					-25	+25	nA/°C	-25	+25	nA/°C

See footnotes at end of device types 01 and 02.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers									Relays closed
				1	2	3	4	5	6	7	8	9	
2 T_A = +125°C (figure 5)	$+I_{IB}$	4001	21	GND			+12.0 V	-6.0 V			-1.0 V		1,2,3,5,6,7,9,10
	$-I_{IB}$	4001	22	GND			+12.0 V	-6.0 V			-1.0 V		1,3,4,5,6,7,9,10
	CMR	4003	23	+5.0 V			+17.0 V	-2.0 V			-6.0 V		2,4,5,6,8,9,10
	CMR	4003	24	-5.0 V			+7.0 V	-12.0 V			+6.0 V		2,4,5,6,8,9,10
	$+I_{CC}$	3005	25	GND		1/	+12.0 V	-6.0 V					1,2,5
	$-I_{CC}$	3005	26	"		"	"	"					1,2,5
	V_{OL}	3007	27	"		"	"	"					1,2,5
	I_{OL}	3009	28	"		"	"	"					1,2,5
	V_{OH1}	3006	29	"	1/		"	"		-5 mA			3,4,6
	V_{OH2}	3006	30	"	"		"	"					3,4,6
	$V_{O(STB)}$ 2/	3007	31	"	"		"	"	300 mV				3,4,6
	$I_{O(STB)}$ 2/	3009	32	"	"		"	"	100 mV				3,4,6
3 T_A = -55°C (figure 5)	V_{IO1}	4001	33	"			"	"			-1.8 V		1,2,3,4,5,6,8,9,10
	V_{IO2}	"	34	"			"	"			-1.8 V		2,4,5,6,8,9,10
	$\Delta V_{IO}/\Delta T$ 3/	"	35	Calculation: $\Delta V_{IO}/\Delta T = \lvert [V_{IO} \text{ (test 33)} - V_{IO} \text{ (test 1)}] \rvert \times 10^3/80°C$									
	I_{IO}	"	36	GND			+12.0 V	-6.0 V			-1.8 V		1,3,5,6,8,9,10
	$\Delta I_{IO}/\Delta T$ 3/	"	37	Calculation: $\Delta I_{IO}/\Delta T = \lvert [I_{IO} \text{ (test 36)} - I_{IO} \text{ (test 3)}] \rvert \times 10^3/80°C$									

See footnotes at end of device types 01 and 02.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Pin measured			Equation	Limits					
								Device type 01			Device type 02		
				No.	Value	Unit		Min	Max	Unit	Min	Max	Unit
2 T_A = +125°C (figure 5)	$+I_{IB}$	4001	21	9	E_{11}	V	$+I_{IB} = E_8 - 10E_{11}$	-.1	+20	µA	-.1	+75	µA
	$-I_{IB}$	4001	22	"	E_{12}	"	$-I_{IB} = 10E_{12} - E_8$	-.1	+20	µA	-.1	+75	µA
	CMR	4003	23	"	E_{13}	"	$CMR = 20 \log \left[\frac{10^3}{E_{13} - E_{14}} \right]$	+80		dB	+80		dB
	CMR	4003	24	"	E_{14}	"							
	$+I_{CC}$	3005	25	4	$+I_{CC}$	mA		+0.5	+9.0	mA	+0.5	+13.5	mA
	$-I_{CC}$	3005	26	5	$-I_{CC}$	mA		-7.0	-0.5	mA	-6.2	-0.5	mA
	V_{OL}	3007	27	7	V_{OL}	V		-1.0		V	-1.0		V
	I_{OL}	3009	28	"	I_{OL}	mA		+0.5		mA	+0.12		mA
	V_{OH1}	3006	29	"	V_{OH1}	V		+2.5	+5.0	V	+2.5	+5.0	V
	V_{OH2}	3006	30	"	V_{OH2}	"		+2.5	+5.0	V	+2.5	+5.0	"
	$V_{O(STB)}$ 2/	3007	31	"	$V_{O(STB)}$	"					-1.0		"
	$I_{O(STB)}$ 2/	3009	32	6	$I_{O(STB)}$	mA					-2.5	-0.1	mA
3 T_A = -55°C (figure 5)	V_{IO1}	4001	33	9	E_{15}	V	$V_{IO} = 10E_{15}$	-3	+3	mV	-4.5	+4.5	mV
	V_{IO2}	"	34	9	E_{16}	V	$V_{IO} = 10E_{16}$	-3	+3	mV	-4.5	+4.5	mV
	$\Delta V_{IO}/\Delta T$ 3/	"	35					-10	+10	µV/°C	-10	+10	µV/°C
	I_{IO}	"	36	9	E_{17}	V	$I_{IO} = E_{15} - E_{17}$	-7	+7	µA	-20	+20	µA
	$\Delta I_{IO}/\Delta T$ 3/	"	37					-75	+75	nA/°C	-75	+75	nA/°C

See footnotes at end of device types 01 and 02.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers									Relays closed
				1	2	3	4	5	6	7	8	9	
3 T_A = -55°C (figure 5)	$+I_{IB}$	4001	38	GND			+12.0 V	-6.0 V			-1.8 V		1,2,3,5,6,7,9,10
	$-I_{IB}$	4001	39	GND			+12.0 V	-6.0 V			-1.8 V		1,3,4,6,7,9,10
	CMR	4003	40	+5.0 V			+17.0 V	-2.0 V			-6.8 V		2,4,5,6,8,9,10
	CMR	4003	41	-5.0 V			+7.0 V	-12.0 V			+3.2 V		2,4,5,6,8,9,10
	$+I_{CC}$	3005	42	GND		1/	+12.0 V	-6.0 V					1,2,5
	$-I_{CC}$	3005	43	"		"	"	"					1,2,5
	V_{OL}	3007	44	"		"	"	"					1,2,5
	I_{OL}	3009	45	"		"	"	"					1,2,5
	V_{OH1}	3006	46	"	1/		"	"		-5 mA			3,4,6
	V_{OH2}	3006	47	"	"		"	"					3,4,6
	$V_{O(STB)}$ 2/	3007	48	"	"		"	"	300 mV				3,4,6
	$I_{O(STB)}$ 2/	3009	49	"	"		"	"	100 mV				3,4,6
4 T_A = +25°C (figure 5)	A_V+	4004	50	"			"	"			-1.9 V		1,2,3,4,5,6,8,9,10
	A_V-	"	51	"			"	"			-0.9 V		1,2,3,4,5,6,8,9,10
5 T_A = +125°C (figure 5)	A_V+	"	52	"			"	"			-1.5 V		1,2,3,4,5,6,8,9,10
	A_V-	"	53	"			"	"			-0.5 V		1,2,3,4,5,6,8,9,10

See footnotes at end of device types 01 and 02.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Pin measured			Equation	Limits					
								Device type 01			Device type 02		
				No.	Value	Unit		Min	Max	Unit	Min	Max	Unit
3 T_A = -55°C (figure 5)	$+I_{IB}$	4001	38	9	E_{18}	V	$+I_{IB} = E_{15} - 10E_{18}$	-.1	+45	µA	-.1	+150	µA
	$-I_{IB}$	4001	39	"	E_{19}	"	$-I_{IB} = 10E_{19} - E_{15}$	-.1	+45	µA	-.1	+150	µA
	CMR	4003	40	"	E_{20}	"	$CMR = 20 \log \left[\frac{10^3}{E_{20} - E_{21}} \right]$	+80		dB	+80		dB
	CMR	4003	41	"	E_{21}	"							
	$+I_{CC}$	3005	42	4	$+I_{CC}$	mA		+0.5	+9.0	mA	+0.5	+13.5	mA
	$-I_{CC}$	3005	43	5	$-I_{CC}$	mA		-7.0	-0.5	mA	-6.2	-0.5	mA
	V_{OL}	3007	44	7	V_{OL}	V		-1.0		V	-1.0		V
	I_{OL}	3009	45	"	I_{OL}	mA		+1.0		mA	+0.25		mA
	V_{OH1}	3006	46	"	V_{OH1}	V		+2.5	+5.0	V	+2.5	+5.0	V
	V_{OH2}	3006	47	"	V_{OH2}	V		+2.5	+5.0	V	+2.5	+5.0	V
	$V_{O(STB)}$ 2/	3007	48	"	$V_{O(STB)}$	V					-1.0		V
	$I_{O(STB)}$ 2/	3009	49	6	$I_{O(STB)}$	mA					-2.5	-0.1	mA
4 T_A = +25°C (figure 5)	A_{V+}	4004	50	9	E_{22}	V	$A_{V+} = 50/(E_1 - E_{22})$	+1,250		V/V	+750		V/V
	A_{V-}	"	51	"	E_{23}	"	$A_{V-} = 50/(E_1 - E_{23})$	+1,250		"	+750		"
5 T_A = +125°C (figure 5)	A_{V+}	"	52	"	E_{24}	"	$A_{V+} = 50/(E_8 - E_{24})$	+1,000		"	+500		"
	A_{V-}	"	53	"	E_{25}	"	$A_{V-} = 50/(E_8 - E_{25})$	+1,000		"	+500		"

See footnotes at end of device types 01 and 02.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers									Relays closed
				1	2	3	4	5	6	7	8	9	
6 T_A = -55°C (figure 5)	A_V+ A_V-	4004 "	54 55	GND "			+12.0 V "	"	-6.0 V		-1.3 V	-2.3 V	1,2,3,4,5,6,8,9,10 1,2,3,4,5,6,8,9,10
7 T_A = +25°C	t_{LTHR} t_{HTHR}		56 57	These tests shall be performed using test conditions and procedures listed on figure 6.									
	t_{SIRL} 2/		58	This test shall be performed using test conditions and procedures listed on figure 7.									

See footnotes at end of device types 01 and 02.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Pin measured			Equation	Limits					
								Device type 01			Device type 02		
				No.	Value	Unit		Min	Max	Unit	Min	Max	Unit
6 T_A = -55°C (figure 5)	A_{V+} A_{V-}	4004 "	54 55	9 9	E_{26} E_{27}	V V	$A_{V^+} = 50/(E_{15} - E_{26})$ $A_{V^-} = 50/(E_{15} - E_{27})$	+1,000 +1,000		V/V "	+500 +500		V/V "
7 T_A = +25°C	t_{LTHR} t_{HTHR}		56 57						60 60	ns ns		60 90	ns "
	t_{STBL} 2/		58									15	"

1/ For device type 01, use -5 mV; for device type 02, use -10 mV.
2/ Device type 02 only.
3/ Test numbers 18, 20, 35, and 37, which require read and record measurements plus a calculation, may be omitted except when subgroups 2 and 3 are being performed for group A sampling inspections and groups B and C end points.
Δ Represents delta.

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TABLE III. Group A inspection for device type 03.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers 1	2	3	4	5	6	7	8	9	10
1 T_A = +25°C (figure 7)	V_{IO1}	4001	1	GND			+12.0 V	-6.0 V			-1.5 V		
	V_{IO2}	"	2	"			"	"			"		
	I_{IO}	"	3	"			"	"			"		
	$+I_{IB}$	"	4	"			"	"			"		
	$-I_{IB}$	"	5	"			"	"			"		
	CMRR	4003	6	+5.0 V			+17.0 V	-2.0 V			-6.5 V		
	CMRR	4003	7	-5.0 V			+7.0 V	-12.0 V			+3.5 V		
	$+I_{CC}$	3005	8	GND		-5 mV	+12.0 V	-6.0 V					
	$-I_{CC}$	3005	9	"		"	"	"					
	V_{OL}	3007	10	"		"	"	"		+100 mA			
	V_{OH}	3006	11	"	-5 mV		"	"		-0.4 mA			
	I_{CEX}		12	"	-5 mV		"	"		+24 V			
	$V_{O(STL)1}$		13	"		-5 mV	"	"	+0.9 V				
	$V_{O(STL)2}$		14	"		"	"	"					+0.9 V
	$V_{O(STH)1}$		15	"		"	"	"	+2.5 V	+16 mA			
	$V_{O(STH)2}$		16	"		"	"	"		+16 mA			+2.5 V
	$I_{STB}1$		17	"		"	"	"	+0.4 V				
	$I_{STB}2$		18	"		"	"	"					+0.4 V
2 T_A = +125°C (figure 7)	V_{IO1}	4001	19	"			"	"			-1.5 V		
	V_{IO2}	"	20	"			"	"			-1.5 V		
	$\Delta V_{IO}/\Delta T$ 1/	"	21	Calculation: $\Delta V_{IO}/\Delta T = \lvert V_{IO}$ (test 19) - V_{IO} (test 1)$\rvert \times 10^3/100°C$									
	I_{IO}	4001	22	GND			+12.0 V	-6.0 V			-1.5 V		
	$\Delta I_{IO}/\Delta T$ 1/	"	23	Calculation: $\Delta I_{IO}/\Delta T = \lvert I_{IO}$ (test 22) - I_{IO} (test 3)$\rvert \times 10^3/100°C$									

See footnote at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Relays energized	Pin measured			Equation	Limits		
					No.	Value	Unit		Min	Max	Unit
1 T_A = +25°C (figure 7)	V_{IO1}	4001	1	1,2,3,4,5,6,8,9,10	9	E_1	V	$V_{IO} = 10E_1$	-2.0	+2.0	mV
	V_{IO2}	"	2	2,4,5,6,8,9,10	"	E_2	"	$V_{IO} = 10E_2$	-2.0	+2.0	mV
	I_{IO}	"	3	1,3,5,6,8,9,10	"	E_3	"	$I_{IO} = E_1 - E_3$	-3.0	+3.0	µA
	$+I_{IB}$	"	4	1,2,3,5,6,7,9,10	"	E_4	"	$+I_{IB} = E_1 - 10E_4$	-.1	+20	µA
	$-I_{IB}$	"	5	1,3,4,5,6,7,9,10	"	E_5	"	$-I_{IB} = 10E_5 - E_1$	-.1	+20	µA
	CMRR	4003	6	2,4,5,6,8,9,10	"	E_6	"	$CMR = 20 \log \frac{10^3}{E_6 - E_7}$	+80		dB
	CMRR	4003	7	2,4,5,6,8,9,10	"	E_7	"				
	$+I_{CC}$	3005	8	1,2,5	4	$+I_{CC}$	mA		+0.5	+10	mA
	$-I_{CC}$	3005	9	1,2,5	5	$-I_{CC}$	mA		-3.6	-0.5	mA
	V_{OL}	3007	10	1,2,5	7	V_{OL}	V			+1.5	V
	V_{OH}	3006	11	3,4,6	"	I_{OL}	V		+2.5	+5.5	V
	I_{CEX}		12	3,4,6	"	I_{CEX}	µA		-.01	+1	µA
	$V_{O(STL)1}$		13	1,2,5	"	$V_{O(STL)1}$	V		+2.5	+5.5	V
	$V_{O(STL)2}$		14	1,2,5	"	$V_{O(STL)2}$	"		+2.5	+5.5	"
	$V_{O(STH)1}$		15	1,2,5	"	$V_{O(STH)1}$	"			+0.4	"
	$V_{O(STH)2}$		16	1,2,5	"	$V_{O(STH)2}$	"			+0.4	"
	I_{STB1}		17	1,2,5	6	I_{STB1}	mA		-3.3	-0.1	mA
	I_{STB2}		18	1,2,5	10	I_{STB2}	mA		-3.3	-0.1	mA
2 T_A = +125°C (figure 7)	V_{IO1}	4001	19	1,2,3,4,5,6,8,9,10	9	E_8	V	$V_{IO} = 10E_8$	-3.0	+3.0	mV
	V_{IO2}	"	20	2,4,5,6,8,9,10	9	E_9	V	$V_{IO} = 10E_9$	-3.0	+3.0	mV
	$\Delta V_{IO}/\Delta T$ 1/	"	21						-10	+10	µV/°C
	I_{IO}	"	22	1,3,5,6,8,9,10	9	E_{10}	V	$I_{IO} = E_8 - E_{10}$	-3.0	+3.0	µA
	$\Delta I_{IO}/\Delta T$ 1/	"	23						-25	+25	nA/°C

See footnote at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers									
				1	2	3	4	5	6	7	8	9	10
2 T_A = +125°C (figure 7)	$+I_{IB}$ $-I_{IB}$	4001 "	24 25	GND GND			+12.0 V +12.0 V	-6.0 V -6.0 V			-1.5 V -1.5 V		
	CMRR CMRR	4003 4003	26 27	+5.0 V -5.0 V			+17.0 V +7.0 V	-2.0 V -12.0 V			-6.5 V +3.5 V		
	$+I_{CC}$ $-I_{CC}$ V_{OL} V_{OH} I_{CEX}	3005 3005 3007 3006	28 29 30 31 32	GND " " " "	 -5 mV -5 mV	-5 mV " " 	+12.0 V " " " "	-6.0 V " " " "	 	 +16 mA -0.4 mA +24 V	 	 	
	$V_{O(STL)1}$ $V_{O(STL)2}$ $V_{O(STH)1}$ $V_{O(STH)2}$		33 34 35 36	" " " "	 	-5 mV " " "	" " " "	" " " "	+0.9 V +2.5 V 	 +16 mA +16 mA	 	 	 +0.9 V +2.5 V
2 T_A = +125°C (figure 5)	I_{STB1} I_{STB2}		37 38	" "	 	" "	" "	" "	+0.4 V 	 	 	 	 +0.4 V

See footnote at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Relays energized	Pin measured			Equation	Limits		
					No.	Value	Unit		Min	Max	Unit
2 T_A = +125°C (figure 7)	$+I_{IB}$ $-I_{IB}$	4001 "	24 25	1,2,3,5,6,7,9,10 1,3,4,5,6,7,9,10	9 "	E_{11} E_{12}	V "	$+I_{IB} = E_8 - 10E_{11}$ $-I_{IB} = 10E_{12} - E_8$		+20 +20	µA µA
	CMRR CMRR	4003 4003	26 27	2,4,5,6,8,9,10 2,4,5,6,8,9,10	" "	E_{13} E_{14}	" "	$CMR = 20 \log \left[\frac{10^3}{E_{13} - E_{14}} \right]$	+80		dB
	$+I_{CC}$ $-I_{CC}$ V_{OL} V_{OH} I_{CEX}	3005 3005 3007 3006	28 29 30 31 32	1,2,5 1,2,5 1,2,5 3,4,6 3,4,6	4 5 7 " "	$+I_{CC}$ $-I_{CC}$ V_{OL} V_{OH} I_{CEX}	mA mA V V µA		+0.5 -3.6 +2.5 -0.5	+10 -0.5 +0.4 +5.5 +100	mA mA V V µA
	$V_{O(STL)1}$ $V_{O(STL)2}$ $V_{O(STH)1}$ $V_{O(STH)2}$		33 34 35 36	1,2,5 1,2,5 1,2,5 1,2,5	" " " "	$V_{O(STL)1}$ $V_{O(STL)2}$ $V_{O(STH)1}$ $V_{O(STH)2}$	V " " "		+2.5 +2.5	+5.5 +5.5 +0.4 +0.4	V " " "
2 T_A = +125°C (figure 5)	I_{STB1} I_{STB2}		37 38	1,2,5 1,2,5	6 10	I_{STB1} I_{STB2}	mA mA		-3.3 -3.3	-0.1 -0.1	mA mA

See footnote at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers									
				1	2	3	4	5	6	7	8	9	10
3 T_A = -55°C (figure 5)	V_{IO1} V_{IO2}	4001 "	39 40	GND GND			+12.0 V +12.0 V	-6.0 V -6.0 V					
	$\Delta V_{IO}/\Delta T$	"	41	Calculation: $\Delta V_{IO}/\Delta T = \lvert V_{IO}$ (test 39) - V_{IO} (test 1)$\rvert \times 10^3/80°C$									
	I_{IO}	"	42	GND			+12.0 V	-6.0 V					
	$\Delta I_{IO}/\Delta T$	"	43	Calculation: $\Delta I_{IO}/\Delta T = \lvert I_{IO}$ (test 42) - I_{IO} (test 3)$\rvert \times 10^3/80°C$									
	$+I_{IB}$ $-I_{IB}$	" "	44 45	GND GND			+12.0 V +12.0 V	-6.0 V -6.0 V					
	CMRR CMRR	4003 4003	46 47	+5.0 V -5.0 V			+17.0 V +7.0 V	-2.0 V -12.0 V					
	$+I_{CC}$ $-I_{CC}$ V_{OL} V_{OH} I_{CEX}	3005 3005 3007 3006	48 49 50 51 52	GND " " " "	 -5 mV -5 mV	-5 mV " " 	+12.0 V " " " "	-6.0 V " " " "		 50 mA -0.4 mA 24 V			
	$V_{O(STL)1}$ $V_{O(STL)2}$ $V_{O(STH)1}$ $V_{O(STH)2}$ I_{STB1} I_{STB2}		53 54 55 56 57 58	" " " " " "		-5 mV " " " " "	" " " " " "	" " " " " "	+0.9 V +2.5 V +0.4 V	 16 mA 16 mA 			 +0.9 V +2.5 V +0.4 V

See footnote at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Relays energized	Pin measured			Equation	Limits		
					No.	Value	Unit		Min	Max	Unit
3 T_A = -55°C (figure 5)	V_{OI1} V_{IO2}	4001 "	39 40	1,2,3,4,5,6,8,9,10 2,4,5,6,8,9,10	9 9	E_{15} E_{16}	V V	$V_{IO} = 10E_{15}$ $V_{IO} = 10E_{16}$	-3 -3	+3 +3	mV mV
	$\Delta V_{IO}/\Delta T$	"	41						-10	+10	µV/°C
	I_{IO}	"	42	1,3,5,6,8,9,10	9	E_{17}	V	$I_{IO} = E_{15} - E_{17}$	-7	+7	µA
	$\Delta I_{IO}/\Delta T$	"	43						-75	+75	nA/°C
	$+I_{IB}$ $-I_{IB}$	" "	44 45	1,2,3,5,6,7,9,10 1,3,4,5,6,7,9,10	9 "	E_{18} E_{19}	V "	$+I_{IB} = E_{15} - 10E_{18}$ $-I_{IB} = 10E_{19} - E_{15}$		+45 +45	µA µA
	CMRR CMRR	4003 4003	46 47	2,4,5,6,8,9,10 2,4,5,6,8,9,10	" "	E_{20} E_{21}	" "	$CMRR = 20 \log \left[\frac{10^3}{E_{20} - E_{21}} \right]$	+80		dB
	$+I_{CC}$ $-I_{CC}$ V_{OL} V_{OH} I_{CEX}	3005 3005 3007 3006	48 49 50 51 52	1,2,5 1,2,5 1,2,5 3,4,6 3,4,6	4 5 7 " "	$+I_{CC}$ $-I_{CC}$ V_{OL} V_{OH} I_{CEX}	mA mA V V µA		+0.5 -3.6 +2.5 -0.5	+10 -0.5 +1.0 +5.5 +100	mA mA V V µA
	$V_{O(STL)1}$ $V_{O(STL)2}$ $V_{O(STH)1}$ $V_{O(STH)2}$ I_{STB1} I_{STB2}		53 54 55 56 57 58	1,2,5 1,2,5 1,2,5 1,2,5 1,2,5 1,2,5	" " " " 6 10	$V_{O(STL)1}$ $V_{O(STL)2}$ $V_{O(STH)1}$ $V_{O(STH)2}$ I_{STB1} I_{STB2}	V " " " mA mA		+2.5 +2.5 -3.3 -3.3	+5.5 +5.5 +0.4 +0.4 -0.1 -0.1	V " " " mA mA

See footnote at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers									
				1	2	3	4	5	6	7	8	9	10
4 T_A = +25°C (figure 5)	A_V+ A_V-	4004 "	59 60	GND "			+12.0 V "	-6.0 V "			-4.5 V -0.5 V		
5 T_A = +125°C (figure 5)	A_V+ A_V-	" "	61 62	" "			" "	" "			-4.5 V -0.5 V		
6 T_A = -55°C (figure 5)	A_V+ A_V-	4004 4004	63 64	" "			" "	" "			-4.5 V -0.5 V		
7 T_A = +25°C	t_{LTHR} t_{HTHR}		65 66	These tests shall be performed using test conditions and procedures listed on figure 6.									
	t_{STRL1} t_{STRL2}		67 68	These tests shall be performed using test conditions and procedures listed on figure 7.									

See footnote at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Relays energized	Pin measured			Equation	Limits		
					No.	Value	Unit		Min	Max	Unit
4 T_A = +25°C (figure 5)	A_{V+} A_{V-}	4004 "	59 60	1,2,3,4,5,6,8,9,10 1,2,3,4,5,6,8,9,10	9 "	E_{22} E_{23}	V "	$A_V^+ = 300/(E_1 - E_{22})$ $A_V^- = 100/(E_{23} - E_1)$	+30 +30		V/V "
5 T_A = +125°C (figure 5)	A_{V+} A_{V-}	" "	61 62	1,2,3,4,5,6,8,9,10 1,2,3,4,5,6,8,9,10	" "	E_{24} E_{25}	" "	$A_V^+ = 300/(E_8 - E_{24})$ $A_V^- = 100/(E_{25} - E_8)$	+10 +10		" "
6 T_A = -55°C (figure 5)	A_{V+} A_{V-}	4004 4004	63 64	1,2,3,4,5,6,8,9,10 1,2,3,4,5,6,8,9,10	" "	E_{26} E_{27}	" "	$A_V^+ = 300/(E_{15} - E_{26})$ $A_V^- = 100/(E_{27} - E_{15})$	+50 +50		" "
7 T_A = +25°C	t_{LTHR} t_{HTHR}		65 66							60 60	ns "
	t_{STRL1} t_{STRL2}		67 68							15 15	" "

1/ Test numbers 21, 23, 41, and 43, which require read and record measurements plus a calculation, may be omitted except when subgroups 2 and 3 are being performed for group A sampling inspections and groups B and C end points.

Δ Represents delta.

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TABLE III. Group A inspection for device types 04 and 05.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers											
				1	2	3	4	5	6	7	8	9	10	11	12
1 T_A = +25°C (figure 5)	V_{IO}	4001	1	GND	GND		+15.0 V	-15.0 V			+15.0 V			GND	+20.0 V
		"	2	"	"		+29.5 V	-0.5 V			+29.5 V			+14.5 V	"
		"	3	"	"		+2.0 V	-28.0 V			+2.0 V			-13.0 V	"
		"	4	"	"		+2.5 V	-2.5 V			+2.5 V			GND	"
	$V_{IO(R)}$	"	5	"	"		+15.0 V	-15.0 V	+15.0 V		+15.0 V			GND	"
		"	6	"	"		+29.5 V	-0.5 V	+29.5 V		+29.5 V			+14.5 V	"
		"	7	"	"		+2.0 V	-28.0 V	+2.0 V		+2.0 V			-13.0 V	"
	I_{IO}	"	8	"	"		+15.0 V	-15.0 V			+15.0 V			GND	"
		"	9	"	"		+29.5 V	-0.5 V			+29.5 V			+14.5 V	"
		"	10	"	"		+2.0 V	-28.0 V			+2.0 V			-13.0 V	"
	$I_{IO(R)}$	"	11	"	"		+15.0 V	-15.0 V	+15.0 V		+15.0 V			GND	"
	$+I_{IB}$	"	12	"	"		+15.0 V	-15.0 V			+15.0 V			GND	"
		"	13	"	"		+29.5 V	-0.5 V			+29.5 V			+14.5 V	"
		"	14	"	"		+2.0 V	-28.0 V			+2.0 V			-13.0 V	"
	$-I_{IB}$	"	15	"	"		+15.0 V	-15.0 V			+15.0 V			GND	"
		"	16	"	"		+29.5 V	-0.5 V			+29.5 V			+14.5 V	"
		"	17	"	"		+2.0 V	-28.0 V			+2.0 V			-13.0 V	"
	$V_{O(STB)}$	"	18	"	+15.0 V		+15.0 V	-15.0 V	-3 mA		+15.0 V				
	$V_{IO(ADJ)+}$	"	19	"	GND		"	"			"			GND	+20.0 V
	$V_{IO(ADJ)-}$	"	20	"	GND		"	"			"			GND	+20.0 V

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter pin numbers 13	Adapter pin numbers 14	Relays energized	Pin measured No.	Pin measured Value	Pin measured Unit	Equation	Limits Min	Limits Max	Limits Unit
1 T_A = +25°C (figure 5)	V_{IO}	4001	1	-20.0 V		1,2,3,4,5	14	E_1	V	$V_{IO} = E_1$	-3.0	+3.0	mV
		"	2	"		1,2,3,4,5	"	E_2	"	$V_{IO} = E_2$	"	"	"
		"	3	"		1,2,3,4,5	"	E_3	"	$V_{IO} = E_3$	"	"	"
		"	4	"		1,2,3,4,5	"	E_4	"	$V_{IO} = E_4$	"	"	"
	$V_{IO(R)}$	"	5	"		1,2,3,4,5,8,9,10	"	E_5	"	$V_{IO(R)} = E_5$	"	"	"
		"	6	"		1,2,3,4,5,8,9,10	"	E_6	"	$V_{IO(R)} = E_6$	"	"	"
		"	7	"		1,2,3,4,5,8,9,10	"	E_7	"	$V_{IO(R)} = E_7$	"	"	"
	I_{IO}	"	8	"		3,4,5	"	E_8	"	$I_{IO} = 20(E_1 - E_8)$	-10.0	+10.0	nA
		"	9	"		3,4,5	"	E_9	"	$I_{IO} = 20(E_2 - E_9)$	"	"	"
		"	10	"		3,4,5	"	E_{10}	"	$I_{IO} = 20(E_3 - E_{10})$	"	"	"
	$I_{IO(R)}$	"	11	"		3,4,5,8,9,10	"	E_{11}	"	$I_{IO} = 20(E_5 - E_{11})$	-25.0	+25.0	"
	$+I_{IB}$	"	12	"		1,3,4,5	"	E_{12}	"	$+I_{IB} = 20(E_1 - E_{12})$	-100	+0.1	"
		"	13	"		1,3,4,5	"	E_{13}	"	$+I_{IB} = 20(E_2 - E_{13})$	-150	"	"
		"	14	"		1,3,4,5	"	E_{14}	"	$+I_{IB} = 20(E_3 - E_{14})$	-150	"	"
	$-I_{IB}$	"	15	"		2,3,4,5	"	E_{15}	"	$-I_{IB} = 20(E_{15} - E_1)$	-100	"	"
		"	16	"		2,3,4,5	"	E_{16}	"	$-I_{IB} = 20(E_{16} - E_2)$	-150	"	"
		"	17	"		2,3,4,5	"	E_{17}	"	$-I_{IB} = 20(E_{17} - E_3)$	-150	"	"
	$V_{O(STB)}$	"	18			1,2,4,5,10	7	E_{18}	"	$V_{O(STB)} = E_{18}$	+14.0		V
	$V_{IO(ADJ)^+}$	"	19	-20.0 V		1,2,3,4,5,8,11	14	E_{19}	"	$V_{IO(ADJ)^+} = E_1 - E_{19}$	+5.0		mV
	$V_{IO(ADJ)^-}$	"	20	-20.0 V		1,2,3,4,5,9,11	14	E_{20}	"	$V_{IO(ADJ)^-} = E_1 - E_{20}$		-5.0	mV

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers											
				1	2	3	4	5	6	7	8	9	10	11	12
1 T_A = +25°C (figure 5)	CMRR	4003	21	Calculation = 20 log $\left[27.5 \times 10^3/(E_3 - E_2)\right]$											
	V_{OL1} 1/	3007	22	+0.5 V	+0.5 V	+6.5 V	+4.5 V	GND		8 mA	GND		GND		
	V_{OL2} 1/	"	23	+3.0 V	+3.0 V	+9.0 V	+4.5 V	GND		8 mA	"		"		
	V_{OL3} 1/	"	24	+13.0 V	+13.0 V	+18.0 V	+15.0 V	-15.0 V		50 mA	"		"		
	V_{OL4} 1/	"	25	-14.0 V	-14.0 V	-9.0 V	+15.0 V	-15.0 V		50 mA	"		"		
	I_{CEX}	3009	26	GND	GND	-5 V	+18.0 V	-18.0 V		+32.0 V	"		"		
	I_{11}	4001	27	+12.0 V	-17.0 V		"	"			"		"		
	I_{12}	4001	28	-17.0 V	+12.0 V		"	"			"		"		
	$+I_{CC}$	3005	29	GND	GND	+10.0 V	+15.0 V	-15.0 V			"		"		
	$-I_{CC}$	3005	30	"	GND	+10.0 V	"	"			"		"		
	I_{OS} 2/	3011	31	"	+0.1 V		"	"		+5.0 V	"		"		

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter pin numbers 13	Adapter pin numbers 14	Relays energized	Pin measured No.	Pin measured Value	Pin measured Unit	Equation	Limits Min	Limits Max	Limits Unit
1 T_A = +25°C (figure 5)	CMRR	4003	21								+80		dB
	V_{OL1} 1/ V_{OL2} 1/ V_{OL3} 1/ V_{OL4} 1/	3007 " " "	22 23 24 25			1,2,12 1,2,12 1,2,12 1,2,12	7 " " "	E_{22} E_{23} E_{24} E_{25}	V " " "	$V_{OL1} = E_{22}$ $V_{OL2} = E_{23}$ $V_{OL3} = E_{24}$ $V_{OL4} = E_{25}$		+0.4 +0.4 +1.5 +1.5	V " " "
	I_{CEX} I_{11} I_{12}	3009 4001 4001	26 27 28			1,2,12 1,2,12 1,2,12	7 1 2	I_{26} I_{27} I_{28}	nA " "	$I_{CEX} = I_{26}$ $I_{11} = I_{27}$ $I_{12} = I_{28}$	-1 -5 -5	+10 +500 +500	nA " "
	$+I_{CC}$ $-I_{CC}$ I_{OS} 2/	3005 3005 3011	29 30 31			1,2 1,2 1,2	4 5 7	I_{29} I_{30} I_{31}	mA " "	$+I_{CC} = I_{29}$ $-I_{CC} = I_{30}$ $I_{OS} = I_{31}$	-5.0	+6.0 +200	mA " "

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers											
				1	2	3	4	5	6	7	8	9	10	11	12
2 T_A = +125°C (figure 5)	V_{IO}	4001	32	GND	GND		+15.0 V	-15.0 V			+15.0 V			GND	+20.0 V
		"	33	"	"		+29.5 V	-0.5 V			+29.5 V			+14.5 V	"
		"	34	"	"		+2.0 V	-28.0 V			+2.0 V			-13.0 V	"
		"	35	"	"		+2.5 V	-2.5 V			+2.5 V			GND	"
	$V_{IO(R)}$	"	36	"	"		+15.0 V	-15.0 V	+15.0 V		+15.0 V			GND	"
		"	37	"	"		+29.5 V	-0.5 V	+29.5 V		+29.5 V			+14.5 V	"
		"	38	"	"		+2.0 V	-28.0 V	+2.0 V		+2.0 V			-13.0 V	"
	$\Delta V_{IO}/\Delta T$	"	39	Calculation = $\boxed{V_{IO}(E_{32}) - V_{IO}(E_1)}$ /100°C 3/											
	I_{IO}	"	40	GND	GND		+15.0 V	-15.0 V			+15.0 V			GND	+20.0 V
		"	41	"	"		+29.5 V	-0.5 V			+29.5 V			+14.5 V	"
		"	42	"	"		+2.0 V	-28.0 V			+2.0 V			-13.0 V	"
	$I_{IO(R)}$	"	43	"	"		+15.0 V	-15.0 V	+15.0 V		+15.0 V			GND	"
	$\Delta I_{IO}/\Delta T$	"	44	Calculation = $\boxed{I_{IO}}$(test 40) - I_{IO}(test 8)$\boxed{\ }$ /100°C 3/											
	$+I_{IB}$	"	45	GND	GND		+15.0 V	-15.0 V			+15.0 V			GND	+20.0 V
		"	46	"	"		+29.5 V	-0.5 V			+29.5 V			+14.5 V	"
		"	47	"	"		+2.0 V	-28.0 V			+2.0 V			-13.0 V	"
	$-I_{IB}$	"	48	"	"		+15.0 V	-15.0 V			+15.0 V			GND	"
		"	49	"	"		+29.5 V	-0.5 V			+29.5 V			+14.5 V	"
		"	50	"	"		+2.0 V	-28.0 V			+2.0 V			-13.0 V	"

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter pin numbers 13	Adapter pin numbers 14	Relays energized	Pin measured No.	Pin measured Value	Pin measured Unit	Equation	Limits Min	Limits Max	Limits Unit
2 T_A = +125°C (figure 5)	V_{IO}	4001	32	-20.0 V		1,2,3,4,5	14	E_{32}	V	$V_{IO} = E_{32}$	-4.0	+4.0	mV
		"	33	"		1,2,3,4,5	"	E_{33}	"	$V_{IO} = E_{33}$	"	"	"
		"	34	"		1,2,3,4,5	"	E_{34}	"	$V_{IO} = E_{34}$	"	"	"
		"	35	"		1,2,3,4,5	"	E_{35}	"	$V_{IO} = E_{35}$	"	"	"
	$V_{IO(R)}$	"	36	"		1,2,3,4,5,8,9,10	"	E_{36}	"	$V_{IO(R)} = E_{36}$	-4.5	+4.5	"
		"	37	"		1,2,3,4,5,8,9,10	"	E_{37}	"	$V_{IO(R)} = E_{37}$	"	"	"
		"	38	"		1,2,3,4,5,8,9,10	"	E_{38}	"	$V_{IO(R)} = E_{38}$	"	"	"
	$\Delta V_{IO}/\Delta T$	"	39								-25.0	+25.0	µV/°C
	I_{IO}	"	40	-20.0 V		3,4,5	14	E_{40}	V	$I_{IO} = 20(E_{40} - E_{32})$	-10.0	+10.0	nA
		"	41	"		3,4,5	"	E_{41}	"	$I_{IO} = 20(E_{41} - E_{33})$	"	"	"
		"	42	"		3,4,5	"	E_{42}	"	$I_{IO} = 20(E_{42} - E_{34})$	"	"	"
	$I_{IO(R)}$	"	43	"		3,4,5,8,9,10	"	E_{43}	"	$I_{IO(R)} = 20(E_{43} - E_{35})$	-25.0	+25.0	"
	$\Delta I_{IO}/\Delta T$	"	44								-100	+100	PA/°C
	$+I_{IB}$	"	45	-20.0 V		1,3,4,5	14	E_{45}	V	$+I_{IB} = 20(E_{32} - E_{45})$	-100	+0.1	nA
		"	46	"		1,3,4,5	"	E_{46}	"	$+I_{IB} = 20(E_{33} - E_{46})$	-150	"	"
		"	47	"		1,3,4,5	"	E_{47}	"	$+I_{IB} = 20(E_{34} - E_{47})$	-150	"	"
	$-I_{IB}$	"	48	"		2,3,4,5	"	E_{48}	"	$-I_{IB} = 20(E_{48} - E_{32})$	-100	"	"
		"	49	"		2,3,4,5	"	E_{49}	"	$-I_{IB} = 20(E_{49} - E_{33})$	-150	"	"
		"	50	"		2,3,4,5	"	E_{50}	"	$-I_{IB} = 20(E_{50} - E_{34})$	-150	"	"

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers											
				1	2	3	4	5	6	7	8	9	10	11	12
2 T_A = +125°C (figure 5)	$V_{O(STB)}$	4001	51	GND	+15.0 V		+15.0 V	-15.0 V	-3 mA		+15.0 V				
	CMRR	4003	52	Calculation = 20 log $\boxed{27.5 \times 10^3/(E_{34} - E_{33})}$											
	V_{OL1} 1/	3007	53	+0.5 V	+0.5 V	+6.5 V	+4.5 V	GND		8 mA	GND		GND		
	V_{OL2} 1/	"	54	+3.0 V	+3.0 V	+9.0 V	+4.5 V	GND		8 mA	"		"		
	V_{OL3} 1/	"	55	+13.0 V	+13.0 V	+18.0 V	+15.0 V	-15.0 V		50 mA	"		"		
	V_{OL4} 1/	"	56	-14.0 V	-14.0 V	-9.0 V	+15.0 V	-15.0 V		50 mA	"		"		
	I_{CEX}	3009	57	GND	GND	-5 V	+18.0 V	-18.0 V		+32.0 V	"		"		
	I_{11}	4001	58	+12.0 V	-17.0 V		"	"			"		"		
	I_{12}	4001	59	-17.0 V	+12.0 V		"	"			"		"		
	$+I_{CC}$	3005	60	GND	GND	+10.0 V	+15.0 V	-15.0 V			"		"		
	$-I_{CC}$	3005	61	"	GND	+10.0 V	"	"			"		"		
	I_{OS} 2/	3011	62	"	+0.1 V		"	"		+5.0 V	"		"		

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter pin numbers 13	Adapter pin numbers 14	Relays energized	Pin measured No.	Pin measured Value	Pin measured Unit	Equation	Limits Min	Limits Max	Limits Unit
2 T_A = +125°C (figure 5)	$V_{O(STB)}$	4001	51			1,2,4,5,10	7	E_{51}	V	$V_{O(STB)} = E_{51}$	+14		V
	CMRR	4003	52								+80		dB
	V_{OL1} 1/ V_{OL2} 1/ V_{OL3} 1/ V_{OL4} 1/	3007 " " "	53 54 55 56			1,2,12 1,2,12 1,2,12 1,2,12	7 " " "	E_{53} E_{54} E_{55} E_{56}	V " " "	$V_{OL1} = E_{53}$ $V_{OL2} = E_{54}$ $V_{OL3} = E_{55}$ $V_{OL4} = E_{56}$		+0.4 +0.4 +1.5 +1.5	V " " "
	I_{CEX} I_{11} I_{12}	3009 4001 4001	57 58 59			1,2 1,2,12 1,2,12	7 1 2	I_{57} I_{58} I_{59}	nA " "	$I_{CEX} = I_{57}$ $I_{11} = I_{58}$ $I_{12} = I_{59}$	-1 -5 -5	+500 " "	nA " "
	$+I_{CC}$ $-I_{CC}$ I_{OS} 2/	3005 3005 3011	60 61 62			1,2 1,2 1,2	4 5 7	I_{60} I_{61} I_{62}	mA " "	$+I_{CC} = I_{60}$ $-I_{CC} = I_{61}$ $I_{OS} = I_{62}$	-5.0	+6.0 +150	mA " "

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers											
				1	2	3	4	5	6	7	8	9	10	11	12
3 T_A = -55°C (figure 5)	V_{IO}	4001 " " "	63 64 65 66	GND " " "	GND " " "		+15.0 V +29.5 V +2.0 V +2.5 V	-15.0 V -0.5 V -28.0 V -2.5 V			+15.0 V +29.5 V +2.0 V +2.5 V			GND +14.5 V -13.0 V GND	+20.0 V " " "
	$V_{IO(R)}$	" " "	67 68 69	" " "	" " "		+15.0 V +29.5 V +2.0 V	-15.0 V -0.5 V -28.0 V	+15.0 V +29.5 V +2.0 V		+15.0 V +29.5 V +2.0 V			GND +14.5 V -13.0 V	" " "
	$\Delta V_{IO}/\Delta T$	"	70	Calculation = $\left[V_{IO}(E_{63}) - V_{IO}(E_1)\right]$ /80°C 3/											
	I_{IO}	" " "	71 72 73	GND " "	GND " "		+15.0 V +29.5 V +2.0 V	-15.0 V -0.5 V -28.0 V			+15.0 V +29.5 V +2.0 V			GND +14.5 V -13.0 V	+20.0 V " "
	$I_{IO(R)}$	"	74	"	"		+15.0 V	-15.0 V	+15.0 V		+15.0 V			GND	"
	$\Delta I_{IO}/\Delta T$	"	75	Calculation = $\left[I_{IO}(\text{test 71}) - I_{IO}(\text{test 8})\right]$ /80°C 3/											
	$+I_{IB}$	" " "	76 77 78	GND " "	GND " "		+15.0 V +29.5 V +2.0 V	-15.0 V -0.5 V -28.0 V			+15.0 V +29.5 V +2.0 V			GND +14.5 V -13.0 V	+20.0 V " "
	$-I_{IB}$	" " "	79 80 81	" " "	" " "		+15.0 V +29.5 V +2.0 V	-15.0 V -0.5 V -28.0 V			+15.0 V +29.5 V +2.0 V			GND +14.5 V -13.0 V	" " "

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter pin numbers 13	Adapter pin numbers 14	Relays energized	Pin measured No.	Pin measured Value	Pin measured Unit	Equation	Limits Min	Limits Max	Limits Unit
3 T_A = -55°C (figure 5)	V_{IO}	4001	63	-20.0 V		1,2,3,4,5	14	E_{63}	V	$V_{IO} = E_{63}$	-4.0	+4.0	mV
		"	64	"		1,2,3,4,5	"	E_{64}	"	$V_{IO} = E_{64}$	"	"	"
		"	65	"		1,2,3,4,5	"	E_{65}	"	$V_{IO} = E_{65}$	"	"	"
		"	66	"		1,2,3,4,5	"	E_{66}	"	$V_{IO} = E_{66}$	"	"	"
	$V_{IO(R)}$	"	67	"		1,2,3,4,5,8,9,10	"	E_{67}	"	$V_{IO(R)} = E_{67}$	-4.5	+4.5	"
		"	68	"		1,2,3,4,5,8,9,10	"	E_{68}	"	$V_{IO(R)} = E_{68}$	"	"	"
		"	69	"		1,2,3,4,5,8,9,10	"	E_{69}	"	$V_{IO(R)} = E_{69}$	"	"	"
	$\Delta V_{IO}/\Delta T$	"	70								-25.0	+25.0	µV/°C
	I_{IO}	"	71	-20.0 V		3,4,5	14	E_{71}	V	$I_{IO} = 20(E_{71} - E_{63})$	-20.0	+20.0	nA
		"	72	"		3,4,5	"	E_{72}	"	$I_{IO} = 20(E_{72} - E_{64})$	"	"	"
		"	73	"		3,4,5	"	E_{73}	"	$I_{IO} = 20(E_{73} - E_{65})$	"	"	"
	$I_{IO(R)}$	"	74	"		3,4,5,8,9,10	"	E_{74}	"	$I_{IO(R)} = 20(E_{74} - E_{67})$	-50.0	+50.0	"
	$\Delta I_{IO}/\Delta T$	"	75								-200.0	+200.0	PA/°C
	$+I_{IB}$	"	76	-20.0 V		1,3,4,5	14	E_{76}	V	$+I_{IB} = 20(E_{63} - E_{76})$	-150	+0.1	nA
		"	77	"		1,3,4,5	"	E_{77}	"	$+I_{IB} = 20(E_{64} - E_{77})$	-200	"	"
		"	78	"		1,3,4,5	"	E_{78}	"	$+I_{IB} = 20(E_{65} - E_{78})$	-200	"	"
	$-I_{IB}$	"	79	"		2,3,4,5	"	E_{79}	"	$-I_{IB} = 20(E_{79} - E_{63})$	-150	"	"
		"	80	"		2,3,4,5	"	E_{80}	"	$-I_{IB} = 20(E_{80} - E_{64})$	-200	"	"
		"	81	"		2,3,4,5	"	E_{81}	"	$-I_{IB} = 20(E_{81} - E_{65})$	-200	"	"

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers											
				1	2	3	4	5	6	7	8	9	10	11	12
3 T_A = -55°C (figure 5)	$V_{O(STB)}$	4001	82	GND	+15.0 V		+15.0 V	-15.0 V	-2 mA		+15.0 V				
	CMRR	4003	83	Calculation = 20 log $\left[27.5 \times 10^3/(E_{65} - E_{64})\right]$											
	V_{OL1} 1/ V_{OL2} 1/ V_{OL3} 1/ V_{OL4} 1/	3007 " " "	84 85 86 87	+0.5 V +3.0 V +13.0 V -14.0 V	+0.5 V +3.0 V +13.0 V -14.0 V	+6.5 V +9.0 V +18.0 V -9.0 V	+4.5 V +4.5 V +15.0 V +15.0 V	GND GND -15.0 V -15.0 V		8 mA 8 mA 50 mA 50 mA	GND " " "		GND " " "		
	I_{I1} I_{I2}	4001 4001	88 89	+12.0 V -17.0 V	-17.0 V +12.0 V		+18.0 V +18.0 V	-18.0 V -18.0 V			" "		" "		
	$+I_{CC}$ $-I_{CC}$ I_{OS} 2/	3005 3005 3011	90 91 92	GND " "	GND GND +0.1 V	+10.0 V +10.0 V	+15.0 V " "	-15.0 V " "		 +5.0 V	" " "		" " "		

See footnotes at the end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter pin numbers 13	Adapter pin numbers 14	Relays energized	Pin measured No.	Pin measured Value	Pin measured Unit	Equation	Limits Min	Limits Max	Limits Unit
3 T_A = -55°C (Figure 5)	$V_{O(STB)}$	4001	82			1,2,4,5,10	7	E_{82}	V	$V_{O(STB)} = E_{82}$	+14		V
	CMRR	4003	83								+80		dB
	V_{OL1} 1/ V_{OL2} 1/ V_{OL3} 1/ V_{OL4} 1/	3007 " " "	84 85 86 87			1,2,12 1,2,12 1,2,12 1,2,12	7 " " "	E_{84} E_{85} E_{86} E_{87}	V " " "	$V_{OL1} = E_{84}$ $V_{OL2} = E_{85}$ $V_{OL3} = E_{86}$ $V_{OL4} = E_{87}$		+0.4 +0.4 +1.5 +1.5	V " " "
	I_{I1} I_{I2}	4001 4001	88 89			1,2,12 1,2,12	1 2	I_{88} I_{89}	nA nA	$I_{I1} = I_{88}$ $I_{I2} = I_{89}$	-5 -5	+500 +500	nA nA
	$+I_{CC}$ $-I_{CC}$ I_{OS} 2/	3005 3005 3011	90 91 92			1,2 1,2 1,2	4 5 7	I_{90} I_{91} I_{92}	mA " "	$+I_{CC} = I_{90}$ $-I_{CC} = I_{91}$ $I_{OS} = I_{92}$	-6.0	+7.0 +250	mA " "

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter PIN numbers											
				1	2	3	4	5	6	7	8	9	10	11	12
4 T_A = +25°C (figure 5)	$A_{VE(REF)}$ A_{VE}^+ A_{VE}^-	4004 " "	93 94 95	GND " "	GND " "		+15.0 V " "	-15.0 V " "		+15.0 V " "		GND +10.0 V -10.0 V			+20.0 V " "
5 T_A = +125°C) (figure 5)	$A_{VE(REF)}$ A_{VE}^+ A_{VE}^-	" " "	96 97 98	" " "	" " "		" " "	" " "		" " "		GND +10.0 V -10.0 V			" " "
6	(Tests 99-101) Same tests, terminal conditions, equations, and limits as subgroup 5 except T_A = -55°C.														
7 T_A = +25°C	t_{RHLC} t_{RLHC}		102 103	These tests shall be performed using test conditions and procedures listed on figure 8.											
8 T_A = +125°C	t_{RHLC} t_{RLHC}		104 105	These tests shall be performed using test conditions and procedures listed on figure 8.											
8 T_A = -55°C	t_{RHLC} t_{RLHC}		106 107	These tests shall be performed using test conditions and procedures listed on figure 8.											

See footnotes at end of device types 04 and 05.

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TABLE III. Group A inspection for device types 04 and 05 - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Adapter pin numbers 13	Adapter pin numbers 14	Relays energized	Pin measured No.	Pin measured Value	Pin measured Unit	Equation	Limits Min	Limits Max	Limits Unit
4 T_A = +25°C (figure 5)	$A_{VE(REF)}$ A_{VE+} A_{VE-}	4004 " "	93 94 95	-20.0 V " "		1,2,3,5,7 1,2,3,5,7 1,2,3,5,7	14 " "	E_{93} E_{94} E_{95}	V " "	$A_{VE}^+ = 10/(E_{94} - E_{93})$ $A_{VE}^- = 10/(E_{93} - E_{95})$	+10.0 +10.0		V/mV "
5 T_A = +125°C (figure 5)	$A_{VE(REF)}$ A_{VE+} A_{VE-}	" " "	96 97 98	" " "		1,2,3,5,7 1,2,3,5,7 1,2,3,5,7	" " "	E_{96} E_{97} E_{98}	" " "	$A_{VE}^+ = 10/(E_{97} - E_{96})$ $A_{VE}^- = 10/(E_{96} - E_{98})$	+8.0 +8.0		" "
6	(Tests 99 - 101) Same tests, terminal conditions, equations, and limits as subgroup 5 except T_A = -55°C.												
7 T_A = +25°C	t_{RHLC} t_{RLHC}		102 103									300 300	ns "
8 T_A = +125°C	t_{RHLC} t_{RLHC}		104 105									500 640	" "
8 T_A = -55°C	t_{RHLC} t_{RLHC}		106 107									300 300	" "

1/ V_{OL1} and V_{OL2} use V_{ID} = -6 mV; V_{OL3}, V_{OL4} use V_{ID} = -5 mV.
2/ Maximum test duration shall be 10 ms.
3/ Test numbers 39, 44, 70, and 75, which require read and record measurements plus a calculation, may be omitted except when subgroups 2 and 3 are being performed for group A sampling inspections and groups B and C end points.
Δ Represents delta.

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TABLE III. Group A inspection for device types 06 and 07.

Subgroup	Symbol	Test no.	Adapter pins 1	2	3	4	5	6	7	Relays energized	Measurement Pin	Value	Unit	Equation	Limit 06 Min	Max	Unit	Limit 07 Min	Max	Unit
1 (Figure 5) T_A = +25°C	V_{IO}	1	15 V	15 V	-15 V	0 V	-1.4 V			3,4,5	10	E_1	V	$V_{IO} = E_1$	-4.0	+4.0	mV	-1	+1	mV
		2	27 V	27 V	-3 V	12 V	-13.4 V			"	"	E_2	"	$V_{IO} = E_2$	"	"	"	"	"	"
		3	3 V	3 V	-27 V	-12 V	10.6 V			"	"	E_3	"	$V_{IO} = E_3$	"	"	"	"	"	"
		4	2.5 V	2.5 V	-2.5 V	-2.5 V	1.1 V			"	"	E_4	"	$V_{IO} = E_4$	"	"	"	"	"	"
	I_{IO}	5	15 V	15 V	-15 V	0 V	-1.4 V			1,2,3,4,5	"	E_5	"	$I_{IO} = 20\ (E_5-E_1)$	-75	+75	nA	-40	+40	nA
		6	27 V	27 V	-3 V	12 V	-13.4 V			"	"	E_6	"	$I_{IO} = 20\ (E_6-E_2)$	"	"	"	"	"	"
		7	3 V	3 V	-27 V	-12 V	10.6 V			"	"	E_7	"	$I_{IO} = 20\ (E_7-E_3)$	"	"	"	"	"	"
	$+I_{IB}$	8	15 V	15 V	-15 V	0 V	-1.4 V			2,3,4,5	"	E_8	"	$+I_{IB} = 20\ (E_1-E_8)$	-0.1	+500	"	-0.1	+500	"
		9	27 V	27 V	-3 V	12 V	-13.4 V			"	"	E_9	"	$+I_{IB} = 20\ (E_2-E_9)$	"	+750	"	"	+750	"
		10	3 V	3 V	-27 V	-12 V	10.6 V			"	"	E_{10}	"	$+I_{IB} = 20\ (E_3-E_{10})$	"	+750	"	"	+750	"
	$-I_{IB}$	11	15 V	15 V	-15 V	0 V	-1.4 V			1,3,4,5	"	E_{11}	"	$-I_{IB} = 20\ (E_{11}-E_1)$	"	+500	"	"	+500	"
		12	27 V	27 V	-3 V	12 V	-13.4 V			"	"	E_{12}	"	$-I_{IB} = 20\ (E_{12}-E_2)$	"	+750	"	"	+750	"
		13	3 V	3 V	-27 V	-12 V	10.6 V			"	"	E_{13}	"	$-I_{IB} = 20\ (E_{13}-E_3)$	"	+750	"	"	+750	"
	CMRR	14	Calculation = $20 \log \left[24 \times 10^3 / \lvert E_3 - E_2 \rvert \right]$												90		dB	90		dB
	V_{OL1}	15	3.5 V		-1 V	-1 V		3.2 mA	7 V	5	8 to 9	E_{14}	V	$V_{OL1} = E_{14}$		+0.4	V		+0.4	V
	V_{OL2}	16	2.25 V		-2.25 V	-2.25 V		3.2 mA	"	"	"	E_{15}	"	$V_{OL2} = E_{15}$		+0.4	"		+0.4	"
	V_{OL3}	17	27 V		-3 V	12 V		25 mA	"	"	"	E_{16}	"	$V_{OL3} = E_{16}$		+1.5	"		+1.5	"
	V_{OL4}	18	3 V		-27 V	-12 V		25 mA	"	"	"	E_{17}	"	$V_{OL4} = E_{17}$		+1.5	"		+1.5	"
	I_{CEX}	19	18 V		-18 V			18 V	-7 V		6	I_1	µA	$I_{CEX} = I_1$	-1.0	+2.0	µA	-1.0	+2.0	µA
	$+I_{CC}$	20	15 V		-15 V				7 V		1	I_2	mA	$+I_{CC} = I_2$		+10.0	mA		+10.0	mA
	$-I_{CC}$	21	15 V		-15 V				"		3	I_3	mA	$-I_{CC} = I_3$	-5.0		mA	-5.0		mA

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TABLE III. Group A inspection for device types 06 and 07 - Continued.

Subgroup	Symbol	Test no.	Adapter pins 1	2	3	4	5	6	7	Relays energized	Measurement Pin	Value	Unit	Equation	Limit 06 Min	Max	Unit	Limit 07 Min	Max	Unit
2 (Figure 5) T_A = +125°C	V_{IO}	22	15 V	15 V	-15 V	0 V	-1.4 V			3,4,5	10	E_{18}	V	$V_{IO} = E_{18}$	-7.0	+7.0	mV	-2.0	+2.0	mV
		23	27 V	27 V	-3 V	12 V	-13.4 V			"	"	E_{19}	"	$V_{IO} = E_{19}$	"	"	"	"	"	"
		24	3 V	3 V	-27 V	-12 V	10.6 V			"	"	E_{20}	"	$V_{IO} = E_{20}$	"	"	"	"	"	"
		25	2.5 V	2.5 V	-2.5 V	-2.5 V	1.1 V			"	"	E_{21}	"	$V_{IO} = E_{21}$	"	"	"	"	"	"
	I_{IO}	26	15 V	15 V	-15 V	0 V	-1.4 V			1,2,3,4,5	10	E_{22}	V	$I_{IO} = 20 (E_{22}-E_{18})$	-75	+75	nA	-40	+40	nA
		27	27 V	27 V	-3 V	12 V	-13.4 V			"	"	E_{23}	"	$I_{IO} = 20 (E_{23}-E_{19})$	"	"	"	"	"	"
		28	3 V	3 V	-27 V	-12 V	10.6 V			"	"	E_{24}	"	$I_{IO} = 20 (E_{24}-E_{20})$	"	"	"	"	"	"
	$+I_{IB}$	29	15 V	15 V	-15 V	0 V	-1.4 V			2,3,4,5	10	E_{25}	V	$+I_{IB} = 20 (E_{18}-E_{25})$	-0.1	+500	nA	-0.1	+500	nA
		30	27 V	27 V	-3 V	12 V	-13.4 V			"	"	E_{26}	"	$+I_{IB} = 20 (E_{19}-E_{26})$	"	+750	"	"	+750	"
		31	3 V	3 V	-27 V	-12 V	10.6 V			"	"	E_{27}	"	$+I_{IB} = 20 (E_{20}-E_{27})$	"	+750	"	"	+750	"
	$-I_{IB}$	32	15 V	15 V	-15 V	0 V	-1.4 V			1,3,4,5	"	E_{28}	"	$-I_{IB} = 20 (E_{28}-E_{18})$	"	+500	"	"	+500	"
		33	27 V	27 V	-3 V	12 V	-13.4 V			"	"	E_{29}	"	$-I_{IB} = 20 (E_{29}-E_{19})$	"	+750	"	"	+750	"
		34	3 V	3 V	-27 V	-12 V	10.6 V			"	"	E_{30}	"	$-I_{IB} = 20 (E_{30}-E_{20})$	"	+750	"	"	+750	"
	CMRR	35	Calculation = 20 Log $\left[\frac{24 \times 10^3}{\lvert E_{24} - E_{23} \rvert}\right]$												90		dB	90		dB
	V_{OL1}	36	3.5 V		-1 V	-1 V		3.2 mA	7 V	5	8 to 9	E_{31}		$V_{OL1} = E_{31}$		+0.4	V		+0.4	V
	V_{OL2}	37	2.25 V		-2.25 V	-2.25 V		3.2 mA	"	"	8 to 9	E_{32}		$V_{OL2} = E_{32}$		+0.4	"		+0.4	"
	V_{OL3}	38	27 V		-3 V	12 V		25 mA	"	"	8 to 9	E_{33}		$V_{OL3} = E_{33}$		+1.5	"		+1.5	"
	V_{OL4}	39	3 V		-27 V	-12 V		25 mA	"	"	8 to 9	E_{34}		$V_{OL4} = E_{34}$		+1.5	"		+1.5	"
	I_{CEX}	40	18 V		-18 V			18 V	-7 V		6	I_4	µA	$I_{CEX} = I_4$	-0.1	+10.0	µA	-1.0	+10.0	µA
	$+I_{CC}$	41	15 V		-15 V				7 V		1	I_5	mA	$+I_{CC} = I_5$		+10.0	mA		+10.0	mA
	$-I_{CC}$	42	15 V		-15 V				"		3	I_6	mA	$-I_{CC} = I_6$	-4.5		mA	-4.5		mA

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TABLE III. Group A inspection for device types 06 and 07 - Continued.

Subgroup	Symbol	Test no.	Adapter pins 1	2	3	4	5	6	7	Relays energized	Measurement Pin	Value	Unit	Equation	Limit 06 Min	Max	Unit	Limit 07 Min	Max	Unit
3 (Figure 6) T_A = -55°C	V_{IO}	43	15 V	15 V	-15 V	0 V	-1.4 V		7 V	3,4,5	10	E_{35}	V	$V_{IO} = E_{35}$	-7.0	+7.0	mV	-2.0	+2.0	mV
		44	27 V	27 V	-3 V	12 V	-13.4 V			"	"	E_{36}	"	$V_{IO} = E_{36}$	"	"	"	"	"	"
		45	3 V	3 V	-27 V	-12 V	10.6 V			"	"	E_{37}	"	$V_{IO} = E_{37}$	"	"	"	"	"	"
		46	2.5 V	2.5 V	-2.5 V	-2.5 V	1.1 V			"	"	E_{38}	"	$V_{IO} = E_{38}$	"	"	"	"	"	"
	I_{IO}	47	15 V	15 V	-15 V	0 V	-1.4 V			1,2,3,4,5	"	E_{39}	V	$I_{IO} = 20 (E_{39}-E_{35})$	-100	+100	nA	-75	+75	nA
		48	27 V	27 V	-3 V	12 V	-13.4 V			"	"	E_{40}	"	$I_{IO} = 20 (E_{40}-E_{36})$	"	"	"	"	"	"
		49	3 V	3 V	-27 V	-12 V	10.6 V			"	"	E_{41}	"	$I_{IO} = 20 (E_{41}-E_{37})$	"	"	"	"	"	"
	$+I_{IB}$	50	15 V	15 V	-15 V	0 V	-1.4 V			2,3,4,5	"	E_{42}	V	$+I_{IB} = 20 (E_{35}-E_{42})$	-0.1	+1000	nA	-0.1	+1000	nA
		51	27 V	27 V	-3 V	12 V	-13.4 V			"	"	E_{43}	"	$+I_{IB} = 20 (E_{36}-E_{43})$	"	"	"	"	"	"
		52	3 V	3 V	-27 V	-12 V	10.6 V			"	"	E_{44}	"	$+I_{IB} = 20 (E_{37}-E_{44})$	"	"	"	"	"	"
	$-I_{IB}$	53	15 V	15 V	-15 V	0 V	-1.4 V			1,3,4,5	"	E_{45}	"	$-I_{IB} = 20 (E_{45}-E_{35})$	"	"	"	"	"	"
		54	27 V	27 V	-3 V	12 V	-13.4 V			"	"	E_{46}	"	$-I_{IB} = 20 (E_{46}-E_{36})$	"	"	"	"	"	"
		55	3 V	3 V	-27 V	-12 V	10.6 V			"	"	E_{47}	"	$-I_{IB} = 20 (E_{47}-E_{37})$	"	"	"	"	"	"
	CMRR	56	Calculation = $20 \text{ Log} \left[24 \times 10^3 / \lvert E_{37} - E_{36} \rvert \right]$												90		dB	90		dB
	V_{OL1}	57	3.5 V		-1 V	-1 V		3.2 mA	7 V	5	8 to 9	E_{48}	V	$V_{OL1} = E_{48}$		+0.6	V		+0.6	V
	V_{OL2}	58	2.25 V		-2.25 V	-2.25 V		3.2 mA	"	"	"	E_{49}	"	$V_{OL2} = E_{49}$		+0.6	"		+0.6	"
	V_{OL3}	59	27 V		-3 V	12 V		25 mA	"	"	"	E_{50}	"	$V_{OL3} = E_{50}$		+1.5	"		+1.5	"
	V_{OL4}	60	3 V		-27 V	-12 V		25 mA	"	"	"	E_{51}	"	$V_{OL4} = E_{51}$		+1.5	"		+1.5	"
	$+I_{CC}$	61	15 V		-15 V				"		1	I_7	mA	$+I_{CC} = I_7$		+11.5	mA		+11.5	mA
	$-I_{CC}$	62	15 V		-15 V				"		3	I_8	mA	$-I_{CC} = I_8$	-6.0		mA	-6.0		mA

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TABLE III. Group A inspection for device types 06 and 07 - Continued.

Subgroup	Symbol	Test no.	Adapter pins 1	2	3	4	5	6	7	Relays energized	Measurement Pin	Value	Unit	Equation	Limit 06 Min	Max	Unit	Limit 07 Min	Max	Unit
4 (Figure 5) T_A = +25°C	A_V	63	15 V	15 V	-15 V		-1.5 V			3,4	10	E_A	V							
			15 V	15 V	-15 V		-11.5 V			"	"	E_{S2}	"	$A_V = 10/(E_A - E_{S2})$	+10		V/mV	+20		V/mV
5 (Figure 5) T_A = +125°C		64	15 V	15 V	-15 V		-1.5 V			3,4	"	E_B	"							
			15 V	15 V	-15 V		-11.5 V			"	"	E_{S3}	"	$A_V = 10/(E_B - E_{S3})$	+5		V/mV	+10		V/mV
6 (Figure 5) T_A = -55°C		65	15 V	15 V	-15 V		-1.5 V			3,4	"	E_C	"							
			15 V	15 V	-15 V		-11.5 V			"	"	E_{S4}	"	$A_V = 10/(E_C - E_{S4})$	+5		V/mV	+10		V/mV

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TABLE III. Group A inspection for device types 06 and 07 - Continued.

Subgroup	Symbol	Test no.	Adapter pins 1	2	3	4	5	6	7	Relays energized	Measurement Pin	Value	Unit	Equation	Limits Min	Max	Unit
9 (Figure 9) T_A = +25°C	V_{IO}		15 V	-15 V	1.4 V		0 V	0 V	5 V	3,4	8	E_O	V				
	Release k31 k4		"	"	"		"	"	"								
	Null V_{IO}		"	"	"	E0	"	"	"								
	Apply 5 mV 00		"	"	"	"	-5 V	"	"								
	Close k1		"	"	"	"	"	"	"	1							
	Apply 100 mV		15 V	-15 V	1.4 V	E0	-5 V	2 mA	5 V	1							
	t_{RLHC}	66	"	"	"	"	"	2 mA	0 V	1	Counter pulse width	t_1	ns	$t_1 = t_{RLHC}$		125	ns
	Reset		"	"	"	"	"	0 V	5 V	1							
	Change 5 mV 00		"	"	"	"	5 V	"	"	1							
	Open k1 close k2		"	"	"	"	"	"	"	2							
	Apply 100 mV		"	"	"	"	"	2 mA	"	2							
	t_{RHLC}	67	"	"	"	"	"	2 mA	0 V	2	Counter pulse width	t_2	ns	$t_2 = t_{RLHC}$		160	ns

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TABLE III. Group A inspection for device types 06 and 07 - Continued.

Subgroup	Symbol	Test no.	Adapter pins							Relays energized	Measurement			Equation	Limits		
			1	2	3	4	5	6	7		Pin	Value	Unit		Min	Max	Unit
10 (Figure 9) T_A = +125°C	V_{IO} *		15 V	-15 V	1.4 V	EO	0 V	0 V	5 V	3,4	8	E_O	V				
	Release k3, k4		"	"	"		"	"	"								
	Null V_{IO}		"	"	"	EO	"	"	"								
	Apply 5 mV OD		15 V	15 V	1.4 V	EO	-5 V	0 V	5 V								
	Close k1		"	"	"	"	"	"	"	1							
	Apply 100 mV		"	"	"	"	"	2 mA	"	1							
	t_{RLHC}	68	15 V	-15 V	1.4 V	EO	-5 V	2 mA	0 V	1	Counter pulse width	t_3	ns	$t_3 = t_{RLHC}$		160	ns
	Reset		"	"	"	"	"	0 V	5 V	1							
	Change 5 mV OD		"	"	"	"	5 V	"	"	1							
	Open k1 Close k2		"	"	"	"	"	"	"	2							
	Apply 100 mV		"	"	"	"	"	2 mA	"	2							
	t_{RHLC}	69	"	"	"	"	"	2 mA	0 V	2	Counter pulse width	t_4	ns	$t_4 = t_{RHLC}$		160	ns

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TABLE III. Group A inspection for device types 06 and 07 - Continued.

Subgroup	Symbol	Test no.	Adapter pins							Relays energized	Measurement			Equation	Limits		
			1	2	3	4	5	6	7		Pin	Value	Unit		Min	Max	Unit
11 (Figure 9) T_A = -55°C	V_{IO}		15 V	-15 V	1.4 V		0 V	0 V	5 V	3,4	8	E_0	V				
	Release k3, k4		"	"	"		"	"	"								
	Null V_{IO}		"	"	"	E0	"	"	"								
	Apply 5 mV 00		"	"	"	"	-5 V	"	"								
	Close k1		"	"	"	"	"	"	"	1							
	Apply 100 mV	70	"	"	"	"	"	2 mA	"	1				$t_5 = t_{RLHC}$		125	ns
	t_{RLHC}		"	"	"	"	"	2 mA	0 V	1	Counter pulse width	t_5	ns				
	Reset		"	"	"	"	"	0 V	5 V	1							
	Change 5 mV 00		"	"	"	"	5 V	"	"	1							
	Open k1 Close k2		"	"	"	"	"	"	"	2							
	Apply 100 mV		"	"	"	"	"	2 mA	"	2							
	t_{RHLC}	71	"	"	"	"	"	2 mA	0 V	2	Counter pulse width	t_6	ns	$t_6 = t_{RHLC}$		160	ns

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TABLE IV. Group C end-point electrical parameters (T_A = +25°C). 1/ 2/ 3/

Test	Device types 01 and 03 $+V_{CC}$ = +12 V, $-V_{CC}$ = -6 V			Device type 02 $+V_{CC}$ = +12 V, $-V_{CC}$ = -6 V			Device types 04 and 05 $+V_{CC}$ = +15 V, $-V_{CC}$ = -15 V			Device type 06 $\pm V_{CC}$ = ±15 V			Device type 07 $\pm V_{CC}$ = ±15 V		
	Limits		Delta	Limits		Delta	Limits		Delta	Limits		Delta	Limits		Delta
	Min	Max		Min	Max		Min	Max		Min	Max		Min	Max	
V_{IO}	-2 mV	+2 mV	±0.5 mV	-3.5 mV	+3.5 mV	±0.5 mV	-3 mV	+3 mV	±0.5 mV	-4 mV	+4 mV	±1 mV	-1 mV	+1 mV	±0.5 mV
$+I_{IB}$	0 µA	20 µA	±2 µA	0 µA	75 µA	±7.5 µA	-100 nA	±0.1 nA	±12.5 nA	-0.1 nA	500 nA	±50 nA	-0.1 nA	+500 nA	±50 nA
$-I_{IB}$	0 µA	20 µA	±2 µA	0 µA	75 µA	±7.5 µA	-100 nA	±0.1 nA	±12.5 nA	-0.1 nA	500 nA	±50 nA	-0.1 nA	+500 nA	±50 nA
I_{CEX} 4/	---	---	---	---	---	---	-1	10 nA	±5 nA	---	---	---	---	---	---

1/ For device types 01, 02, and 03, V_{IO} deltas and limits apply to tests 1 and 2 in table III. For device types 04 and 05, V_{IO} deltas and limits apply to tests 1, 2, and 3 in table III. For device types 06 and 07, V_{IO} deltas and limits apply to tests 1, 2, and 3, in table III.

2/ For device types 01, 02, and 03, $+I_{IB}$ deltas and limits apply to test 4 in table III. For device types 04 and 05, $+I_{IB}$ deltas and limits apply to tests 12, 13, and 14 in table III. For device types 06 and 07, $+I_{IB}$ deltas and limits apply to tests 8, 9, and 10, in table III.

3/ For device types 01, 02, and 03, $-I_{IB}$ deltas and limits apply to test 5 in table III. For device types 04 and 05, $-I_{IB}$ deltas and limits apply to tests 15, 16, and 17 in table III. For device types 06 and 07, $-I_{IB}$ deltas and limits apply to tests 11, 12, and 13, in table III.

4/ I_{CEX} deltas and limits apply to test 26 in table III.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

6.2 Acquisition data. The acquisition document must specify the following:

a. Complete PIN (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to acquiring activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths or lead forming, if applicable. These requirements shall not affect the PIN.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510 and MIL-STD-1331 and as follows.

6.3.1 Logic threshold voltage. The approximate voltage at the output of the comparator at which the loading logic circuitry changes its digital state.

6.3.2 Voltage gain. The ratio of the change in output voltage to the change in voltage between the input terminals producing it with the dc output level in the vicinity of the logic threshold voltage (A_V).

6.3.3 Response time. The interval between the application of an input step function and the time when the output crosses the logic threshold voltage. The input step drives the comparator from some initial, saturated input voltage to an input level just barely in excess of that required to bring the output from saturation to the logic threshold voltage. This excess is referred to as the voltage overdrive.

6.3.4 Positive output level. The dc output voltage in the positive direction with the input voltage equal to or greater than a minimum specified amount.

6.3.5 Negative output level. The dc output voltage in the negative direction with the input voltage equal to or greater than a minimum specified amount.

6.3.6 Output sink current. The maximum negative current that can be delivered by the comparator (I_{OL}).

6.3.7 Peak output current. The maximum current that may flow into the output load without causing damage to the comparator.

6.3.8 Differential input voltage. The difference between the two voltages applied to the input terminals of an amplifier. The difference is considered positive when the noninverting input is positive with respect to the inverting input and negative when the inverting input is positive with respect to the noninverting input (V_{ID}).

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6.3.9 Strobe current. The current into the strobe terminal of an amplifier (I_{STB}).

6.3.10 Output leakage current. The current into the output of an amplifier with the output at high level (I_O).

6.3.11 Ground leakage current. The current into the ground terminal of an amplifier with the output at high level (I_G).

6.3.12 Input leakage current. The current into the input terminal of an amplifier with the output at a specified voltage (I_I).

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired to Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the PIN.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	710
02	711
03	LM106
04	LM111
05	LM2111
06	LM119
07	LT119A

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - MC
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, OS, SH

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-M-38510/103E	2. DOCUMENT DATE *(YYMMDD)*

3. DOCUMENT TITLE Military Specification, Microcircuits, Linear, Voltage Comparators, Monolithic Silicon

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER	
a. NAME	b. ORGANIZATION
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*
	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY	
a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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